

TI Solar Charged Battery Pack

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FINAL REPORT

REVISION – 2
28 April 2025

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CONCEPT OF OPERATIONS

REVISION – 2

28 April 2025

CONCEPT OF OPERATIONS
FOR
TI Solar Powered Battery Pack

TEAM 64

APPROVED BY:

Patrick Wang Date

Prof. Kalafatis Date

Ali Alinezi Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	9/10/2024	TI Solar Powered Battery Pack	Patrick Wang	Draft Release
2	9/22/2024	TI Solar Powered Battery Pack	Patrick Wang	Revision 1

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1. Executive Summary

The purpose of this project is to design a battery pack capable of charging phones and other small electronic devices with solar panels and USB Type C input to provide a convenient and eco-friendly charging experience to users without the constraints of user access to power outlets. The battery pack will be able to take USB C and solar inputs, which will deliver 15 Watts to the battery. With the subsystems of solar panel array, battery charger, battery gauge, batteries, power multiplexer (power MUX), and microcontroller, the battery pack will be able to provide charge to other 5V 3A devices with a Type C output, utilizing maximum power point tracking (MPPT) algorithm to optimize charging efficiency. The microcontroller unit will connect and control different subsystems, providing the charging feature and several operation modes such as charging mode, power supply mode, low-power mode, sleep/idle mode, and safety mode with the LED indicator. The integrated system will provide a better charging experience to users in various use cases, from everyday use to special cases like camping and emergency scenarios.

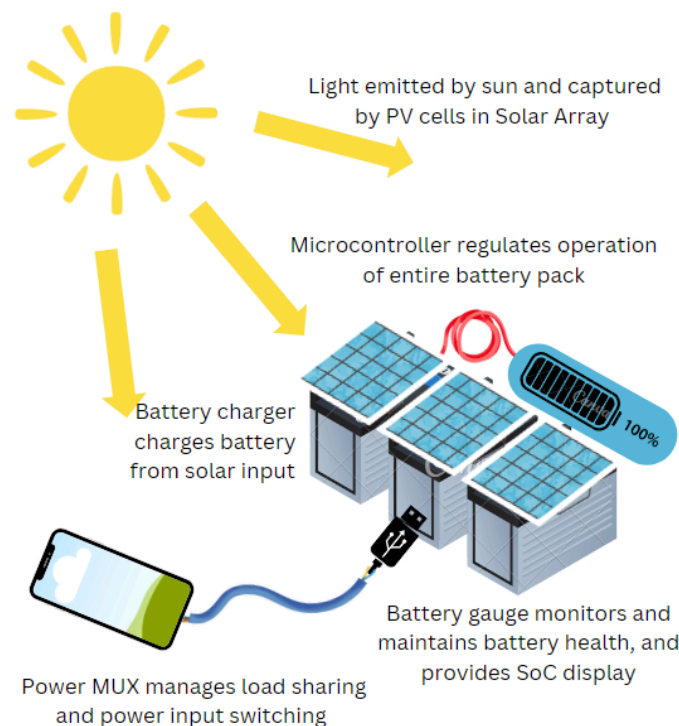


Figure 1: Project Conceptual Image of Proposed Battery Pack System

2. Introduction

This document serves as an introduction to TI Solar Battery Pack, an electronic system that charges portable electronic devices such as smartphones or calculators, and is charged via either a solar panel or USB-C port. To meet increasing needs for eco-friendly energy sources and provide the ability to charge regardless of whether users have access to electricity outlets or not, the battery pack will have the option to charge itself with the attached solar panel.

2.1. Background

Since mobile electronic devices like smartphones have become an important part of daily life, demand for portable chargers or power banks has also increased. Many people carry portable chargers to charge their small portable devices, but the power banks themselves also require battery charge from power outlets. Our proposed project, TI Solar Charged Battery Pack, would enhance currently-existing power banks by providing an additional charging option: charging via the solar panel attached. This new charging option will let users charge their power banks wherever they are located, even when they don't have access to power outlets they can use. Moreover, the option to charge the battery pack using solar energy will help reduce carbon footprint so that the users' everyday lives become more eco-friendly.

2.2. Overview

The system, or the solar battery pack, will consist of two inputs, a solar panel and power delivered via USB-C, one USB-C output, a power MUX to choose one of the inputs at the time of charging, a battery charger, a battery pack, a battery gauge, a SOC display to inform the battery pack status to users, and a microcontroller unit, or MCU to control all the subsystems within the battery pack system. To achieve the goal of designing a battery pack for charging phones or other small electronics, the IC chips produced by TI (BQ2538 1S battery charger, BQ27z746 1S battery gauge and protector, MSPM0 MCU, TPS25814 Type-C port controller, TMUX1248 with back to back TPS2595 eFuses for bidirectional power control) will be used and assembled on a PCB designed by each subsystem owners to integrate different subsystems into one whole system. Also, to protect the inner parts of the battery pack and prevent direct contact with PCB, the system will be enclosed by a 3D-printed enclosure. As a whole, the final deliverable will be a working prototype of the solar charging battery pack consisting of various PCB boards, IC chips, and other components enclosed by a 3D-printed enclosure. The characterization and functionality of the complete system will be tested under different conditions to ensure safety for correct functionality.

2.3. Referenced Documents and Standards

- MSPM0L130x Mixed-Signal Microcontrollers datasheet (Rev. D)
<https://www.ti.com/lit/ds/symlink/mspm0l1306.pdf?ts=1724842475793>

- MSPM0 Design Flow Guide (Rev. C)
https://www.ti.com/lit/ug/slaaed1c/slaaed1c.pdf?ts=1724793262284&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FMSPM0L1306
- MSPM0 L-Series 32-MHz Microcontrollers Technical Reference Manual (Rev. D)
<https://www.ti.com/lit/pdf/slau847>
- Achieve Bidirectional Control and Protection Through Back-to-Back Connected eFuse <https://www.ti.com/jp/lit/an/slva948/slva948.pdf?ts=1726079227330>
- TPS2595xx 2.7 V to 18 V, 4-A, 34-mΩ eFuse With Fast Overvoltage Protection datasheet (Rev. C)
<https://www.ti.com/lit/ds/symlink/tps2595.pdf?src-supplier=IHS+Markit>
- TMUX1248 3-Ω Low RON, 5-V, 2:1 (SPDT) General Purpose Switch with 1.8-V Logic datasheet
https://www.ti.com/lit/ds/symlink/tmux1248.pdf?ts=1726376111023&ref_url=https%253A%252F%252Fwww.google.com%252F
- BQ27Z746 Impedance Track™ Technology Battery Gas Gauge and Protection Solution for 1-Series Cell Li-Ion Battery Packs
<https://www.ti.com/lit/gpn/bq27z746>
- EV2400 Evaluation Module Interface Board User's Guide. (Rev. D)
https://ti.com/lit/ug/sluiu446d/sluiu446d.pdf?ts=1726440596938&ref_url=https%253A%252F%252Fwww.ti.com%252Ftool%252FEV2400

3. Operating Concept

3.1. Scope

The task of this project is to create a solar battery pack that can effectively capture the sun's energy, store the energy in a power bank, and use the stored power to charge electronic devices whenever needed. In terms of power conversion, storage, and delivery, the project will concentrate on the right solar panel to use, suitable battery chargers and gauges, and compatible MUX and microcontroller systems. The battery pack will ideally be portable, easy to operate, and flexible in its use for charging different electronics in situations where there is no access to electricity. The project will comprise the evaluation of the system's feasibility, effectiveness, and practicality, and may potentially find applicability for other purposes beyond our capstone project.

3.2. Operational Description and Constraints

The portable solar battery pack will serve as an independent power source for use in off-grid homes or other applications like during camping or any other sporting activities that require power supply, when electric power is not available. The system operates using photovoltaic PV cells by converting solar energy to electrical energy and storing it in rechargeable 18650 batteries. The intended use of operations for this battery pack is for people to be able to charge any of their electronic devices, ranging from mobile phones and small tablets to medical equipment and anything else that relies on a USB port connection.

Some key operation constraints include the availability of sunlight, as it directly influences the charging efficiency and capability of the unit. Another potential constraint could be the form factor if portability is a goal for this project. It can be a constraint to build a small, light, and ergonomic battery pack unit with potential material cost constraints. Other potential constraints may include material durability and environmental factors such as temperature changes and water resistance. More constraints could involve ensuring the unit avoids overheating or overcharging during extensive use.

3.3. System Description

The Solar battery pack can be broken down to the following subsystems which are critical to the functionality of the system.

- **Solar Panel Array:** This sub system depends on the solar energy and converts this energy to electrical energy. The array is made up of photovoltaic cells which produce DC electricity when placed under natural sunlight. This subsystem efficiency depends on the type of solar cells, orientation of the panels used and the climate condition of the particular area.
- **Battery Charger:** Another sub system of the design is the battery charger that has the capability to control the amount of electrical current that is used by the battery from the solar panel. In addition, it also avoids the batteries being over charged and the circuit controls the voltage and the current that charges the batteries. The

charger is also required for MPPT calculation with an aim of maximizing energy that can be taken from the panel.

- **Battery Gauge:** The battery gauge gives the users a real-time estimate of the amount of charge remaining in the battery pack. The interface will be a form of Light Emitting Diodes popularly referred to as LED. This particular system is intended to help users understand how much charge is left for their devices.
- **Batteries:** The converted energy from the solar panels will be stored in the batteries. We will be using Lithium-ion batteries because of their high energy density, long cycle life, and inherent safety. Battery capacity selection is done according to the energy storage capacity desired with a compromise between the unit weight and energy storage capability. The selected batteries will also be compatible with the battery gauge and chargers with appropriate chemistry IDs and programming to ensure that the gauge meets compatibility requirements and is able to read the health of the batteries.
- **Power Multiplexer (Power MUX):** The power mux is the unit that regulates power sharing between the batteries, the solar panel and the connected appliances. It regulates how power is provided from one of these constituents to the other in order to supply power to devices, as well as controlling the charging of batteries and taking in energy from the solar panels. This subsystem can have power management priority, which means that energy can be passed based on the needs and according to the existing supply.
- **Microcontroller:** The microcontroller controls and supervises all the other subsystems of the system hence taking the central role of a control unit. From the sensors, battery status, solar input, it takes data about the charges, discharges, and distributes power. In safety features, it manages voltage monitors, controlling of power mux and checking the optimum usage of energy. This may also enable interfacing with other external devices for monitoring or data acquisition.

These subsystems work together to produce not just a working proof of concept, but an actual practical, efficient, and portable solar battery charger that can always help keep electronics charged.

3.4. Modes of Operations

The solar-powered battery pack operates in several modes:

- **Charging Mode:** The system charges the batteries using solar energy. It stops charging when the battery is full.
- **Power Supply Mode:** This mode would be when the battery charges devices connected to the system from the battery.
- **Low-Power Mode:** In low-power mode, the unit will alert the user via the state-of-charge display or LEDs that it is in low on power.
- **Sleep/Idle Mode:** The closed system shuts down when offline, saving power by turning off the state of charge display or LEDs until a device connects or the unit is charging via solar input.
- **Safety Mode:** When a fault is engaged, for instance due to overcharging or overheating, the system restricts operations to prevent damage to the components.

3.5. *Users*

The solar power battery pack is suitable for all customers but may be more beneficial to those living in rural areas or with no access to electricity. Other customers may include hikers, mountain climbers, and rescue teams. The solar battery pack will be an easy-to-operate system that will not require much training to manage it. The user should not need any more than an elementary understanding of how to connect electronics to power sources. All features, including the output ports and battery gauges, will be user-friendly.

The ones to benefit most will be those who require portable and renewable energy, which may include hikers, campers, disaster management teams, and those in the rural areas where there is likely to be little power supply. The system is also perfect for consumers who are concerned with their environmental impacts and wish to decrease their dependency on conventional power stations.

3.6. *Support*

Given that the unit we are making is a simple, functional battery pack, there will only be a user manual explaining the basic operations of the battery pack unit. Specifications of the batteries and other technical aspects will be included in the manual, but the manual itself will contain all of the support for the user.

4. Scenario(s)

4.1. *Camping*

One of the main issues while camping is keeping devices charged. Even with portable batteries, there's still a limit on how much power is available, and the trade-off with high-capacity battery packs is their heavy weight. The solar-charged battery pack solves this by being able to take a solar input to recharge during the day, then, once charged, disconnect from the solar panel and be a light extra charge for any USB-C device. This solution solves the problem of hauling a large and heavy battery and provides a renewable energy source during the trip.

4.2. *Emergency Scenarios*

During emergency scenarios, time and power are limited. With a high-power solar input, the battery can be quickly charged in response to a disaster scenario. Once the battery is charged, the device uses a maximum power point tracking (MPPT) algorithm to charge the device at maximum speed with the highest possible efficiency, providing a fast and efficient charge when time and resources are at a minimum.

5. Analysis

5.1. *Summary of Proposed Improvements*

- The system provides a flexible solution for charging the battery pack, which will be done using USB-C or solar panels.
- The MCU will use MPPT technology to ensure the maximum power transfer rate between battery and the device, given a wide range of weather and operating conditions.
- The battery bank will monitor the battery's State of Health (SOH) to prevent further degradation and maintain the quality state of the battery.
- Three 5W solar panels will be used in parallel to deliver up to 15W of power to charge the battery. This will allow for a smaller form factor and more flexibility in the physical device itself.
- Implementing these changes will allow users to charge small devices such as a phone in both areas with and without access to power from an outlet.

5.2. *Disadvantages and Limitations*

- The efficiency of the solar panel charging will be largely dependent on the environment that the panel is placed in. The panel will be limited in environments such as at night, cloudy days, or inside facilities.
- The size and portability of a solar panel may be more bulky and less convenient than battery packs without a solar option.
- The device may degrade faster due to the weather's impact from leaving the solar panel and battery pack outside. Such factors include rain, extreme hot or cold temperatures, and other particulates in the air.

5.3. *Alternatives*

Here is a list of alternative solutions to the proposed solar battery pack solution

- A simple USB-C wall charger will provide reliable charging if there is an accessible outlet and reliable power to that outlet. This solution is not portable and will not work in remote off-grid locations.
- A custom solar grid with a battery bank will provide a greater and more robust solar option, but it will be much larger and not as easily portable.
- A portable battery charger without solar compatibility is an alternative solution that will also provide portable charging to small devices as long as the battery pack is properly charged. However, this solution does nothing to generate power. Thus, the charge available to a user on a trip solely depends on the battery's charge before use.

5.4. *Impact*

This system will positively impact the environment by providing a clean, renewable energy source that can charge phones and other small electronic devices. The everyday person will

be able to contribute to a lower carbon footprint by introducing a solar method of charging many devices that they use on a daily basis. Additionally, many areas that are either remote or lack the proper infrastructure to support reliable power will be able to utilize this system to ensure that many essential devices get charged. The main environmental concern for the system will come from proper handling and disposal of lithium-ion battery packs that are potentially harmful if not properly disposed of.

TI Solar Charged Battery Pack

Jaein Cha

Cole Miller

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Patrick Wang

FUNCTIONAL SYSTEM REQUIREMENTS

REVISION – 2

28 April 2025

FUNCTIONAL SYSTEM REQUIREMENTS

FOR

TI Solar Powered Battery Pack

PREPARED BY:

Team 64	Date
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APPROVED BY:

Patrick Wang	Date
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John Lusher, P.E.	Date
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Ali Alinezi	Date
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Change Record

Rev.	Date	Originator	Approvals	Description
1	9/10/2024	TI Solar Powered Battery Pack	Patrick Wang	Draft Release
2	9/22/2024	TI Solar Powered Battery Pack	Patrick Wang	Revision 1

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1. Introduction

1.1. Purpose and Scope

This project aims to create a battery pack capable of charging mobile phones amongst other portable electronic devices using solar panels and a USB Type C port. This charging alternative will be useful to people who cannot rely on the availability of power outlets. The battery pack will be able to take USB C and solar inputs, which will deliver 15 Watts to the battery. The battery pack features 4 main subsystems: the battery charger, battery gauge, power multiplexer, and the microcontroller. These subsystems work cohesively to supply a 5V 3A charge to other devices through a Type C output. The microcontroller unit will manage the various subsystems, enabling the charging capability and other operational modes, such as charging mode, power supply mode, and low power mode. The integrated system provides a practical means of charging devices in environments with unreliable power sources.

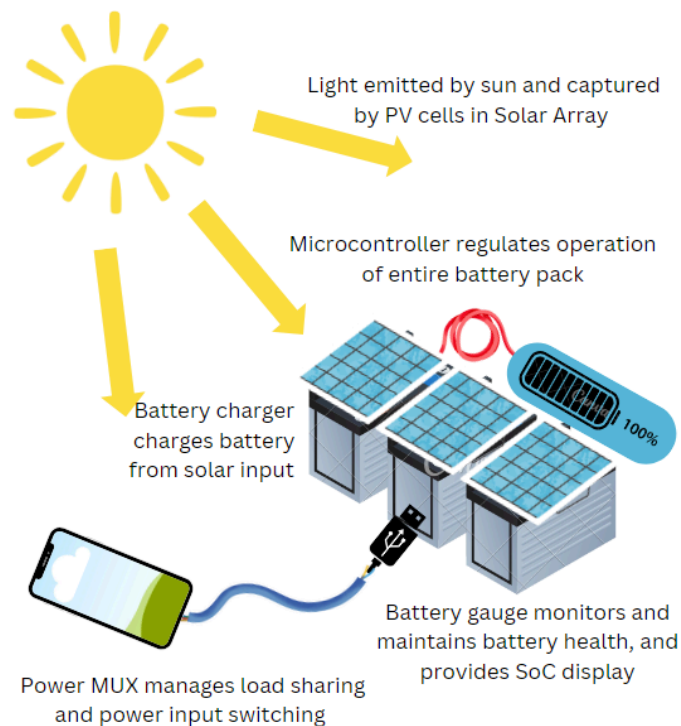


Figure 1: Project Conceptual Image of Proposed Battery Pack System

1.2. Responsibility and Change Authority

The team leader is Patrick Wang who will be responsible for verifying that all requirements of the project are met and completed. These requirements can only be changed with the team leader's approval, Professor Stavros Kalafatis, and Professor John Lusher.

Subsystem	Member Name
Microcontroller	Jaein Cha
Battery Gauge	Patrick Wang
Battery Charger	Cole Miller
Power Multiplexer	Camden Pomeroy

Table 1: Subsystem Partition

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

Document Number	Revision/ Release Date	Document Title
IEEE 370-2020	01/08/2021	IEEE Standard for Electrical Characterization of Printed Circuit Board and Related Interconnects at Frequencies up to 50 GHz
IEEE 2030.2.1-2019	12/13/2019	IEEE Guide for Design, Operation, and Maintenance of Battery Energy Storage Systems, both Stationary and Mobile, and Applications Integrated with Electric Power Systems
IEC 61730	Edition 3.0 - 09/13/2023	Photovoltaic (PV) Module Safety Qualification
UM10204	Rev. 7.0 — 10/1/2021	I2C-bus specification and user manual
IEC 62680-1-3:2022	Edition 5.0 - 09/05/2022	Environmental Engineering Considerations and Laboratory Tests

Table 2: Applicable Documents

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

Document Number	Revision/Release Date	Document Title
SLUSF18A	Rev. A - 12/04/2021	BQ25638I2C Controlled, 5-A, Maximum 18-V Input, Charger with NVDC Power Path Management and USB OTG Boost Output datasheet (Rev. A)
SLUSDW2A	Rev. A - 12/08/2021	BQ27Z746 Impedance Track Technology Battery Gas Gauge and

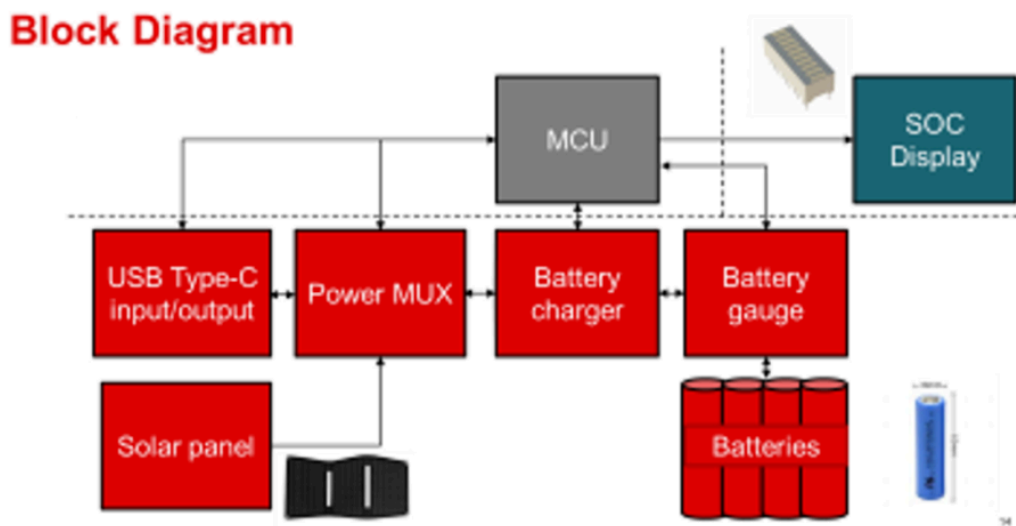
		Protection Solution for 1-Series Cell Li-Ion Battery Packs datasheet (Rev. A)
SCDS447	Rev. A 06/01/2021	TMUX1248 3-Ω Low RON, 5-V, 2:1 (SPDT) General Purpose Switch with 1.8-V Logic datasheet
SLLSEN9F	Re. F 03/15/2022	TUSB320 USB Type-C Configuration Channel Logic and Port Control datasheet
SLVSGA8B	Rev. B - 04/04/2022	TPS25946xx 2.7-V–23-V, 5.5-A, 28-mΩ eFuse with Bidirectional Current Support
None listed	R7E - 03-29-2023	Voltaic Systems P105

Table 3: Reference Documents

2.3. Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings or other documents that are invoked as “applicable” in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICDs that have their relevant documents considered to be incorporated as cited.



Power Mux: The power multiplexer is responsible for switching between multiple power sources to supply stable power to the electronic device it is charging. It is also responsible for managing the power load between multiple devices, while also making sure the loads all receive a stable power supply. This ensures safe and reliable handling of charging and discharging of the battery.

3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Extreme Temperature Operation

The unit shall be functional in extreme temperatures, from -18°C to 55°C (0°F to 131°F).

Rationale: This range allows the user to be able to take their battery pack to go hiking in extreme conditions with extreme temperatures, and they can still rely on the battery pack to charge their phones. This usage case can also apply to other users who may be stuck in the midst of a natural disaster, which may entail extreme weather conditions where they will need a reliable power source to power their devices.

3.2.1.2. Usable Battery Time

The unit shall not have a self-discharge rate greater than 5% per 24 hours.

Rationale: The battery needs to be able to hold charge so it can provide charge when needed. If it self-discharges all of its power within a day, the battery pack is not practical. Hence, the battery pack needs to have a small and slow self-discharge rate to ensure it is useful and practical when needed.

3.2.1.3. Solar Charging Time

The solar power charging system shall be able to provide enough power to run a maximum of 12 hours irrespective of weather conditions.

Rationale: There are approximately 12 ± 2 hours a day of sunlight, but the consistency of solar rays and energy conversion is not always guaranteed constant, so the battery pack should be able to provide enough power to charge during times when solar charging is suboptimal.

3.2.2. Physical Characteristics

3.2.2.1. Mass

The mass of the Solar Battery Pack shall be less than 30 pounds (13.6 kilograms)

Rationale: This is a requirement because the battery pack cannot be too heavy. If it is too heavy, it may be difficult to carry, and would hence be impractical for hikers during their adventures.

3.2.2.2. Volume Envelope

The volume envelope of the Solar Battery Pack shall be less than or equal to 1.25 inches in height, 5.85 inches in width, and 8.80 inches in length.

Rationale: The battery pack needs to have a practical form factor that allows it to be carried easily, or else it becomes more of a hindrance than a useful tool for users who need to use the battery pack but can't move large objects easily.

3.2.2.3. Material Composition

The battery pack shall be dust resistant, made from quality materials that ensure corrosion resistance and durability.

Rationale: The battery pack should have a quality casing and internal components because especially during times of potential danger, the last thing a user would want is for their power supply to fall apart in their hands. Hence, the battery pack needs to be made of quality materials that ensure durability in extreme conditions.

3.2.3. Electrical Characteristics

3.2.3.1. Inputs

The presence or absence of any combination of the input signals in accordance with ICD specifications applied in any sequence shall not damage the Solar Battery Pack System, reduce its life expectancy, or cause any malfunction, either when the unit is powered or when it is not.

No sequence of command shall damage the Solar Battery Pack System, reduce its life expectancy, or cause any malfunction.

Rationale: By design, the unit should limit the chance of damage or malfunction by user/technician error.

3.2.3.1.1 Power Consumption

The maximum peak power of the system shall consume 1.5 watts.

Rationale: This is a requirement specified by our customer as the mandatory charge efficiency rating.

3.2.3.1.2 Input Voltage Level

The input voltage level for the Battery Pack shall be +3 VDC to +18 VDC.

Rationale: That is the range that the battery charger can accept.

3.2.3.1.3 External Commands

The Solar Battery Pack system shall document all external commands in the appropriate ICD.

Rationale: The ICD will capture all interface details from the low level electrical to the high-level packet format.

3.2.3.2. Outputs

3.2.3.2.1 State of Charge Output

The Battery Pack shall include a state of charge display.

Rationale: The Battery Pack must communicate the current charge level of the device.

3.2.3.2.2 Voltage Output

The Battery pack shall output +3VDC to +18VDC.

Rationale: Those are the predefined limits that the charger can output.

3.2.3.2.3 Diagnostic Output

The Solar Battery Pack System may include a diagnostic interface for error logging.

Rationale: Provides the ability to see any errors that may occur from charging through specific blinking patterns of the LEDs.

3.2.4. Environmental Requirements

The Solar Battery Pack System shall be designed to withstand and operate in the environments and laboratory tests specified in the following section.

Rationale: The Battery Pack will need to function in a variety of extreme environment settings.

3.2.4.1. Pressure (Altitude)

The Solar Battery Pack shall be able to function properly at altitudes ranging from sea level to 400 feet above sea level

Rationale: The Battery Pack will need to function in a variety of extreme altitude conditions.

3.2.4.2. Thermal

The Solar Battery Pack may be able to function properly in an environment with temperatures ranging from -18°C to 55°C (0°F to 131°F).

Rationale: The Battery Pack will need to function in a variety of extreme thermal conditions.

3.2.4.3. Humidity

The Solar Battery Pack may be able to function properly in an environment with relative humidity ranging from 0% to 100%.

Rationale: The Battery Pack will need to function in a variety of extreme humidity conditions.

3.2.5. Failure Propagation

The Solar Battery Pack system may have failure detection in the form of a system check whenever it is in its “wake” state. When the unit is powered on and detects a connected device, it will send power to the device, and if it receives any error when sending power, it will alert the microcontroller and battery gauge of the error, and will relay to the SOC display that there is an error during startup through the form of blinking LEDs.

3.2.5.1. Failure Detection, Isolation, and Recovery (FDIR)

3.2.5.1.1 Built In Test (BIT)

The Solar Battery Pack System may have an internal subsystem that will generate test signals and evaluate the Battery Pack System responses and determine if there is a failure.

3.2.5.1.1.1 BIT Critical Fault Detection

The Solar Battery Pack shall be able to detect a critical fault in regular operation 100 percent of the time.

Rationale: This requirement would allow the Battery Pack system to detect critical faults and inform the user.

3.2.5.1.1.2 BIT False Alarms

The Battery Pack System shall have a BIT false alarm rate of less than 5 percent.

Rationale: This would limit the number of false alerts that the user has to check.

3.2.5.1.1.3 BIT Log

The BIT may save the results of each test to a log that shall be stored in the Battery Pack MCU in case of troubleshooting.

Rationale: This allows users to monitor and diagnose past trends in errors and can help in troubleshooting.

3.2.5.1.2 Isolation and Recovery

The Solar Battery Pack System should provide for fault isolation and recovery by enabling subsystems to be reset or disabled based upon the result of the BIT.

Rationale: In the case of extreme system errors, the Battery Pack system may need to reset certain subsystems to allow the unit to function properly.

4. Support Requirements

The Solar Battery Pack system will come with a manual on how to connect the solar panels to the unit. Regarding support for the unit, there will be an FAQ section in the manual to help with troubleshooting. The manual will include information on all of the subsystems and how they work, so if the unit is not functioning properly, the customer would be able to test each subsystem individually to troubleshoot and identify any problems, and resort to the FAQ in the manual to fix any errors.

Appendix A: Acronyms and Abbreviations

BIT	Built-In Test
CCA	Circuit Card Assembly
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EO/IR	Electro-optical Infrared
FOR	Field of Regard
FOV	Field of View
Hz	Hertz
ICD	Interface Control Document
IEC	International Electric Code
IEEE	Institute of Electrical and Electronics Engineers
kHz	Kilohertz (1,000 Hz)
LED	Light-emitting Diode
mA	Milliamp
MHz	Megahertz (1,000,000 Hz)
MPPT	Maximum Power Point Tracking
MTBF	Mean Time Between Failure
MTTR	Mean Time To Repair
MUX	Multiplexer
mW	Milliwatt
PCB	Printed Circuit Board
PMUX	Power Multiplexer
RMS	Root Mean Square
TBD	To Be Determined
TTL	Transistor-Transistor Logic
USB	Universal Serial Bus
MCU	Microcontroller Unit
I2C	Inter-Integrated Circuit

Appendix B: Definition of Terms

Maximum Power Point Tracking - A control algorithm to maximize power extraction.

Appendix C: Interface Control Document

The interface Control Document is attached as a separate document.

TI Solar Charged Battery Pack

Jaein Cha

Cole Miller

Camden Pomeroy

Patrick Wang

INTERFACE CONTROL DOCUMENT

REVISION – 2

28 April 2025

INTERFACE CONTROL DOCUMENT

FOR

TI Solar Charged Battery Pack

PREPARED BY:

Team 64	Date
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APPROVED BY:

Patrick Wang	Date
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John Lusher, P.E.	Date
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Ali Alinezi	Date
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Change Record

Rev.	Date	Originator	Approvals	Description
1	9/10/2024	TI Solar Powered Battery Pack	Patrick Wang	Draft Release
2	9/22/2024	TI Solar Powered Battery Pack	Patrick Wang	Revision 1

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1. Executive Summary

This ICD will serve as the system control standard, which will cover each subsystem of the TI Solar Charged Battery pack. This document will detail the battery charger, battery gauge, power mux, and MCU subsystem as well as the communication between them. Specifications and rationale will be given for each interface, and protocol.

2. References and Definitions

2.1. References

Document Number	Revision/Release Date	Document Title
IEEE 370-2020	01/08/2021	IEEE Standard for Electrical Characterization of Printed Circuit Board and Related Interconnects at Frequencies up to 50 GHz
IEEE 2030.2.1-2019	12/13/2019	IEEE Guide for Design, Operation, and Maintenance of Battery Energy Storage Systems, both Stationary and Mobile, and Applications Integrated with Electric Power Systems
IEC 61730	Edition 3.0 - 09/13/2023	Photovoltaic (PV) Module Safety Qualification
UM10204	Rev. 7.0 — 10/1/2021	I2C-bus specification and user manual
IEC 62680-1-3:2022	Edition 5.0 - 09/05/2022	Environmental Engineering Considerations and Laboratory Tests

Table 1: Reference Documents

2.2. Definitions

CCA	Circuit Card Assembly
mA	Milliamp
mW	Milliwatt
MHz	Megahertz (1,000,000 Hz)
TBD	To Be Determined
TTL	Transistor-Transistor Logic
VME	VERSA-Module Europe
MCU	Microcontroller Unit
I2C	Inter-Integrated Circuit

3. Physical Interface

3.1. *Weight*

The mass of the Solar Battery Pack shall be less than 30 pounds (13.6 kilograms). The solar panels make up 1.25 pounds. While fully charged the batteries will weigh around 0.33 pounds. The case of the battery pack should be around .75 pounds. The rest of the components' weights should be negligible.

3.2. *Dimensions*

3.2.1. Dimension of Solar Panel

The solar panels' dimensions are 5.827 x 8.78 x 0.2 inches.

3.2.2. Dimension of Battery

The battery's dimensions are 0.75 inches in diameter and 2.7 inches in length.

4. Thermal Interface

4.1. *Battery Cooling*

In order to help keep the battery's temperature in a safe range of operation, the side of the case not mounting the USB-C connections will be a mesh pattern and not fully closed. This will accelerate cooling and allow hot air to leave the case. To help speed up the cool-down process, heatsinks will be attached to the parts that generate the most heat during initial testing along a cold wall.

5. Electrical Interface

5.1. *Power MUX*

There are two primary inputs of power; one input is a 15W solar panel and the other is 15W USB C charging. The nominal voltage and amperage are 5V and 3A. The power flow will be through the inputs and sent to the Vbus of the battery charger. Each line has a TPS 25946 eFuse that will handle overvoltage, undervoltage, and bidirectional power flow. A signal communicated from the MCU to a TMUX 1248 will select the proper power path, and ensure that there will be no short circuits. This is performed in a break before making configuration, which only allows one input to charge the battery at a time.

5.2. Battery Charger

The charger will receive 15 watts of power through VBUS. Then communicating with the battery gauge will step down the voltage using a buck converter to charge the battery through the battery gauge, and provide the necessary 5V 3A if not already supplied. Once a sink is added to VBUS the charger will receive a signal, and increase the batteries output voltage using a boost converter and supply it to VBUS. All DC-DC converters are based on solid state devices. The battery charger requires 4 GPIO pins for clock, interrupt, enable, and sending data.

5.3. Battery Gauge

The BQ27z746 battery gauge is used to track and regulate the voltage of the batteries, specifically for single lithium-ion and lithium-polymer batteries. It communicates with a microcontroller through an I2C (SMBus) pin for data in, out and error indicating pins. This chip has a connection point right into the positive terminal of the battery to monitor the battery's voltage, while there are sensors for temperature read by thermistors connected on pins. It operates on voltage range 2.5 V – 4.5 V supplied from the battery and connected to the microcontroller. It also contains general purpose I/O (GPIO) that can be configured for any function including controlling the status of LEDs or outside circuits. There is also an alert pin to indicate if the battery is low or too hot for normal operation. It can also control external MOSFET switches for the protection of the battery during charging and discharging stages.

5.4. Battery Pack

The battery pack consisting of three Samsung INR18650 batteries connected in series will produce a total voltage of about 10.8V, or 3.6V for each battery. It has a similar capacity in milliamps hours (mAh) to a standard INR18650 cell. There are two electrical interfaces at the series end mainly, the positive terminal through which a device is powered or the pack is charged, and the negative terminal. When batteries are connected in series, the voltage rises and the pack can charge high voltage loads but the overall energy capacity in mAh remains the same. The battery gauge will protect against any problems related to the battery such as over charge, over-discharge, and short circuit during the operation of the battery pack.

5.5. User Control Interface

The battery charger will have a SOC display and a button as part of the user control interfaces. Users will have information regarding the state of charge of the battery, current mode of operation, and error codes when something exceptional happens that prevents the battery pack from normal operations. To let users give direct inputs to the battery pack, the battery pack will have a button so the user can turn on/off the device charging operation, reset the system, and switch between different modes of operation. The SOC display will be made of 5 LEDs connected to the 5-bit bus, which is connected to the MCU's one of the GPIO pins. 5-bit signals generated from the connected GPIO pin will be decomposed into 5 1-bit signals to light up or turn off each LEDs. The button will be connected to one of the GPIO pins of MCU so the microcontroller can take user inputs from the button.

5.6. *Microcontroller Unit*

The microcontroller unit (MCU) will receive a supply voltage of 3.5V (supported range is between 1.6V to 3.6V). It has GPIO pins up to 24 out of 28 pins total for enabling electrical connection to different subsystems for clock signals, data transfer, interrupt, power management, and more. There are two GPIOs that are 5-V tolerant and two of them that are high-drive. The core has a frequency up to 80 MHz.

6. Communications / Device Interface Protocols

6.1. *I2C Communication*

Communication between all systems will be handled by the I2C communication method. This will operate according to the specifications provided by NXP in UM10204.

6.2. *USB C Port Interface*

The input port will use USB Type C, and TUSB320 for Type C configuration channel logic and port control. The MCU will use I2C communication to send the proper signals through CC1 and CC2.

6.3. *Solar Panel Interface*

The Solar panel interface by a Waterproof Male 3.5x1.1mm and connect to a port on the side

TI Solar Charged Battery Pack

Jaein Cha

Cole Miller

Camden Pomeroy

Patrick Wang

SCHEDULE AND VALIDATION

REVISION – 2
28 April 2025

SCHEDULE AND VALIDATION
FOR
TI Solar Charged Battery Pack

PREPARED BY:

Team 64 Date

APPROVED BY:

Patrick Wang Date

John Lusher, P.E. Date

Ali Alinezi Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	9/10/2024	TI Solar Powered Battery Pack	Patrick Wang	Draft Release
2	9/22/2024	TI Solar Powered Battery Pack	Patrick Wang	Revision 1
3	4/28/2025	TI Solar Powered Battery Pack	Patrick Wang	Revision 2

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1. Schedule

	Deadline	Started	Done
Subsystem Research	8/26/24	Yes	Yes
Subsystem Design	9/2/24	Yes	Yes
ConOps Report	9/9/24	Yes	Yes
MCU Development Environment Setup	9/9/24	Yes	Yes
MCU Basic Exemplary Code Practice	9/9/24	Yes	Yes
Battery Selection	9/9/24	Yes	Yes
SOC Display selection	9/9/24	Yes	Yes
Solar Panel selection	9/16/24	Yes	Yes
Receiving EVMs and parts	9/16/24	Yes	Yes
Schematic Design (All subsystems except MCU)	9/16/24	Yes	Yes
Schematic Calculations (All subsystems except MCU)	9/23/24	Yes	Yes
Order Solar Panel	9/23/24	Yes	Yes
Order batteries	9/23/24	Yes	Yes
Schematic Design (MCU)	9/23/24	Yes	Yes
PCB Layout Design	10/7/24	Yes	Yes
PCB Prototyping/Assembly	10/14/24	Yes	Yes
MCU Firmware development (I2C) *	10/28/24	Yes	No
Validation Testing	11/25/24	Yes	Yes
Subsystem finish	1/31/2025	Yes	Yes
Start integration of charger and gauge	2/15/2025	Yes	Yes
Start integration of MUX + charger + gauge	2/15/2025	Yes	Yes
Start integration of MCU with all subsystems	2/15/2025	Yes	Yes
Design integrated PCB with all subsystems	2/28/2025	Yes	Yes
Design 3D enclosure	3/15/2025	Yes	Yes
Integrate and test TUSB subsystem into board	3/23/2025	Yes	Yes
Validate all tests on new board	3/30/2025	Yes	Yes

Table 1: Schedule

* MCU Firmware development (I2C) - the primary development is done, but further validation of I2C behavior correctness needs to be done to conclude the development process is actually done, because it is not done unless the program is 100% validated not to have any bugs. The validation couldn't be done correctly because of a connection problem on the board.

2. Validation Plan

Number	Validation Test:	Input:	Expected Output:	Results	Status	Engineer
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	4.5V of Vout when input was 5V		Camden Pomeroy
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected		Camden Pomeroy
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is Low	As expected		Camden Pomeroy
4	PP1 PG (Bad signal)	if Vin < 2V	PG signal is High	As expected		Camden Pomeroy
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is High	As expected		Camden Pomeroy
6	PP2 PG (Bad signal)	if Vin < 2V	PG signal is Low	As expected		Camden Pomeroy
7	PP1 UVLO Operational	if Vin < UVLO	No output	As expected		Camden Pomeroy
8	PP2 UVLO Operational	if Vin < UVLO	No output	As expected		Camden Pomeroy
9	PP1 OVLO Operational	if Vin > OVLO	No output	As expected		Camden Pomeroy
10	PP2 OVLO Operational	if Vin > OVLO	No output	As expected		Camden Pomeroy
11	SEL PP1	SEL is high	PP1 is active and PP2 is not	Sel pin would not properly switch		Camden Pomeroy
12	SEL PP2	SEL is low	PP2 is active and PP1 is not	Sel pin would not properly switch		Camden Pomeroy
13	ILM during startup	Inrush current of 4A	Limited current (<4 A) to ramp up	Could not get DC load to operate		Camden Pomeroy

14	Separate Power Paths	S1 active and both Vins 5V	Only PP1 Vout = 5V	As expected		Camden Pomeroy
15	Seperate Power Paths	S2 active and both Vins 5V	Only PP2 Vout = 5V	As expected		Camden Pomeroy
16	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected		Camden Pomeroy

Number	Validation Test:	Input:	Expected Output:	Results	Status	Engineer
1	Powered IC - 5V 3A	5V 3A	5V 3A	As expected		Patrick Wang
2	IC valid I2C connection	IC connect ed to EV2400	IC communicates with bqstudio	As expected		Patrick Wang
3	accurate voltage measurements in bqstudio	3V to 5V	3V to 5V	As expected		Patrick Wang
4	accurate battery capacity and charge level measurements	plugged into 100% power source	100%	As expected		Patrick Wang
5	accurate temperature sensing measurements	21.8 degrees C	21.8 degrees C	As expected		Patrick Wang
6	IC is on latest firmware update	IC flashed to latest firmware	IC updated	As expected		Patrick Wang
7	IC BAT+ charging voltage of 3.7V	3.7V into BAT+	3.7V into VC2	As expected		Patrick Wang
8	IC PACK+ input voltage 3.5 to 5.5V	CHG and DSG match Vin	CHG and DSG match Vin	they match indicating FETs open		Patrick Wang

9	IC PACK+ input voltage outside of range of 3.5 to 5.5V	CHG and DSG should be 0	CHG and DSG match Vin	to indicate FETs off		Patrick Wang
10	IC PACK+ Vin powers IC at same voltage	3V to 5V	3V to 5V	As expected		Patrick Wang

Number	Validation Test:	Input:	Expected Output:	Results	Status	Engineer
1	Continuity test - before soldering	Continuity test using DMM	Beep (connected) for same nets, no sound otherwise	As expected		Jaein Cha
2	Continuity test - after soldering	Continuity test using DMM	Beep (connected) for same nets, no sound otherwise	As expected		Jaein Cha
3	Voltage/current test (MCU)	3.3V to the power pin (VCC)	3.3V from the source GPIO pins	As expected (from SCL pin)		Jaein Cha
4	Voltage/current test (TUSB)	3.3V to the power pin (VCC)	3.3V from the source GPIO pins	As expected (from ID pin)		Jaein Cha
5	I2C - voltage level	3.3V to the power pin (VCC)	Same voltage of 3.3V for both ends of I2C bus	As expected		Jaein Cha
6	I2C - initialization	3.3V Power supply to the power pin (VCC)	Read TUSB I2C address	TUSB unresponsive		Jaein Cha
7	I2C - receive (DFP connected)	DFP plug is connect	Read TUSB I2C address and value, blinks LEDs corresponding to	TUSB unresponsive		Jaein Cha

		ed to the port	the code for "DFP inserted"			
8	I2C - receive (UFP connected)	UFP plug is connected to the port	Read TUSB I2C address and value, blinks LEDs corresponding to the code for "UFP inserted"	TUSB unresponsive		Jaein Cha
9	I2C - receive (DRP connected)	DRP plug is connected to the port	Read TUSB I2C address and value, blinks LEDs corresponding to the code for "UFP inserted"	TUSB unresponsive		Jaein Cha
10	Reset	Reset button pressed	MCU resets and stop working while the button is pressed	Yes on LP, not on custom board		Jaein Cha
11	SOP Display	SOP button	Lights up all the LEDs when SOP button is pressed	Yes on LP, not on custom board		Jaein Cha

Number	Validation Test:	Input:	Expected Output:	Results	Status	Engineer
1	Standard Operation	5V 3A on VBUS	4.2 V and 90% power efficiency	3.7V 2A on VBAT		Cole Miller
2	USB OTG Function	4.2V 3.2A on VBAT	around 4.2V 3.2A on VBAT	5V 1.5A on VBUS		Cole Miller
3	ILIM Register Functionality	If input current >1A	output current >1A	Output limited to 2A since register can't be accessed		Cole Miller
4	VLIM Functionality	If Vin > 4.2V	Vo > 4.2V	Output limited to 3.7V		Cole Miller
5	PG Pin Functionality	Good input	Low signal	Low signal		Cole Miller
6	STAT Pin Functionality	Charging	Low signal	Low signal		Cole Miller

7	STAT Pin Functionality	Not Charging	High	High		Cole Miller
8	System Overvoltage Protection	Vin > Vsys_op	Vin > Vsys_op	Vsys_fault_flag = 1		Cole Miller
9	ILIM Hardware Functionality	Input current > 3A	Input current > 3A	Current limited to 3A		Cole Miller
10	IC valid I2C connection	IC connected to EV2400	IC communicates with bqstudio	doesn't communicate		Cole Miller
11	Solar Panel VOC	Solar panel Voc < 18V	Solar panel Voc < 18V	As expected		Cole Miller
12	Solar Panel Power	regular usage in sunny day	output 15W	As expected		Cole Miller
13	ILIM Pin Disable	Able to disable automatic current limit using register	Able to disable automatic current limit using register	Unable to control		Cole Miller
14	Switching Speed	1.5 MHz on SWS pin	1.5 MHz on SWS pin	1.5 MHz on SWS pin		Cole Miller

Table 2: Validation plan

3. Performance on Execution Plan

The execution plan was executed completely, if not to the best of our ability. Most of the tasks were able to be completed independently of the other subsystems in terms of equipment usage and validation testing. All of the achievable goals in this first semester were met with all of the objectives outlined at the beginning of the project being demonstrated by the initial proposed solution. In regards to the class outcomes, the team

completed and satisfied the goals of the course in a timely manner, with new skills in designing, fabricating, assembling, and testing PCBs.

4. Performance on Validation Plan

The validation tests were run to the best of the team's ability, with a few minor setbacks. Most of these setbacks are due to soldering issues, or known issues in the layout design of the PCB. Data was collected on the different subsystems to diagnose the issues and solve them preemptively before integration next semester. Although some issues popped up, each team member has a general idea of how to fix the issues and make their subsystems work as desired. The only issue is that the team ran out of time to actually implement such changes because some of them required ordering new boards.

More details on each subsystem's individual validation tests will be covered in each of their respective documents in this report.

TI Solar Charged Battery Pack

Jaein Cha

Cole Miller

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SUBSYSTEM REPORT

REVISION – 1

28 April 2025

SUBSYSTEM REPORT
FOR
TI Solar Powered Battery Pack

TEAM 64

APPROVED BY:

Patrick Wang Date

Prof. Kalafatis Date

Ali Alinezi Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	12/3/2024	TI Solar Powered Battery Pack	Patrick Wang	Revision 1

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1. Introduction

The Solar Charger Battery Pack is a highly efficient battery pack designed to work with both USB-C and solar inputs. The system takes 15 watts from either USB-C or solar inputs and through a switch mode buck converter charges a 1S3P battery pack. Then when charging a device, it detects a sink connected to USB-C and supplies 15 watts of power. The system is broken down into multiple subsystems including USB-C input, solar panel, power mux, MCU, batteries, and battery gauge.

2. Battery Gauge - Patrick Wang

2.1. Subsystem Introduction

The battery gauge is a key element in the battery pack system that monitors the overall performance of the battery pack. The BQ28z620 accurately measures parameters such as state of charge, input and output voltage, and temperature to protect against extreme outliers by shutting off the system to prevent damage. This is important, as prohibiting overcharging and discharging can help extend the lifetime of batteries.

2.2. Subsystem Details

The designs for the schematic and layout were heavily influenced by the BQ28z620 datasheet and EVM layout. The diagrams are illustrated below.

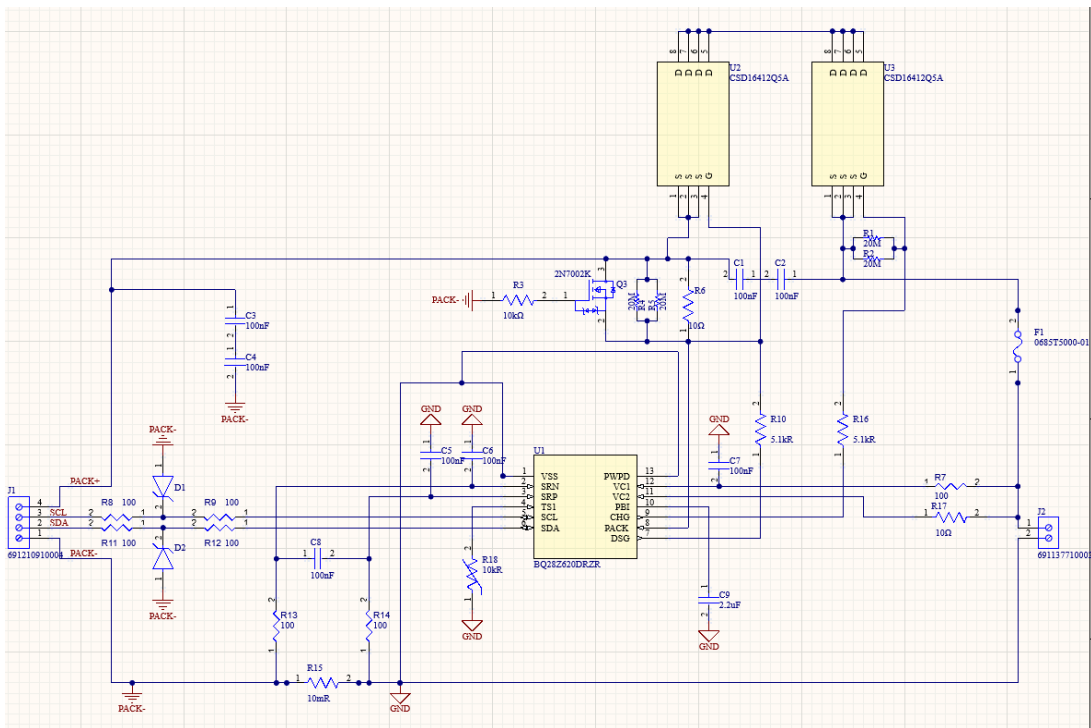


Figure 1: Battery Gauge Schematic

A lot of the design decisions came from the schematic on the BQ28z620 datasheet, but also from another BQ27z746 IC that I worked with before. Such examples include connecting sense resistors SRN and SRP to capacitors close to the IC, and having them connected parallel to both the sense resistor R15 and capacitor C8. Also including TVS diodes on the SDA and SCL pins as shown in other datasheets looked like a great idea to protect against transient voltage spikes, so those were added to the schematic.

Some values were calculated, but most of the capacitor values were already given in the datasheet that was designed to work with a similar 1S3P battery cell configuration. I chose to implement a higher 5A instead of the recommended 3A to account for potentially higher input current from the battery charger, as it may supply varying inputs as high as 5.5A that the battery gauge needs to be able to handle before tripping.

The layout design is illustrated below.

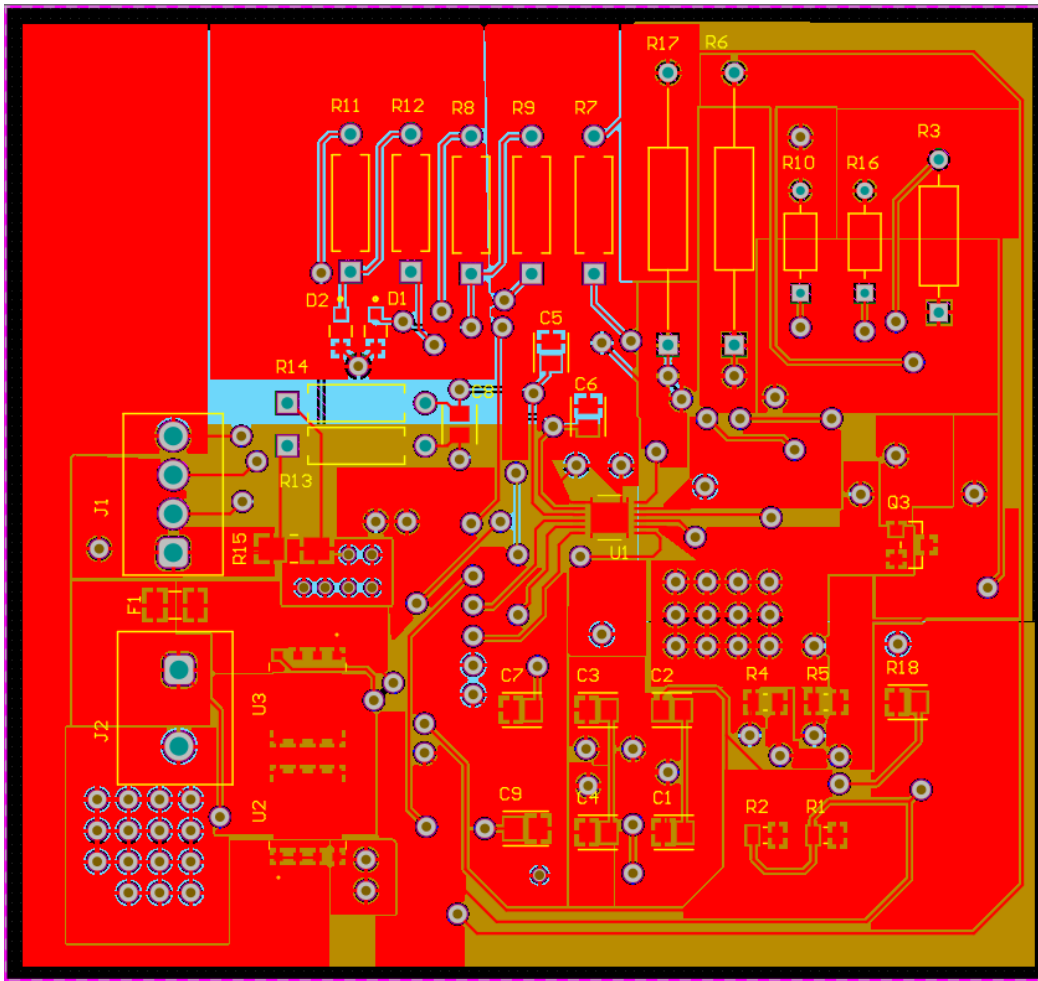


Figure 2: Battery Gauge Layout

Many design factors had to be taken into consideration when creating this layout. The datasheet specified that capacitors C5 and C6 needed to be placed near the IC, and my sponsor also told me the Kelvin Connection between the sense resistor pins were very sensitive to noise, so the path between the SRN and SRP pins to the sense resistor R15 had to be as short and direct as possible.

The layout is intended to be as neat as possible, with through-hole resistors grouped in one area, surface mount resistors grouped in another area, capacitors grouped together, MOSFETs U1 and U2 placed right next to each other, and the connectors grouped together as well. The Kelvin connection involving C5, C6, R13, R14, and R15 are intended to create a quick path to feed the SRN and SRP analog I/O to the resistor.

Being a four layer board, this PCB implements a lot of tight groups of vias to help with better signal routing and layer transition, as well as improving the power distribution and heat dissipation. The big poly pours around every pin that involves a power path is also intended to help with power dissipation.

2.3. Subsystem Validation

A general image of the validation tests run is shown below.

Number	Validation Test:	Input:	Expected Output:	Results	Status	Engineer
1	Powered IC - 5V 3A	5V 3A	5V 3A	As expected		Patrick Wang
2	IC valid I2C connection	IC connected to EV2400	IC communicates with bqstudio	As expected		Patrick Wang
3	accurate voltage measurements in bqstudio	3V to 5V	3V to 5V	As expected		Patrick Wang
4	accurate battery capacity and charge level measurements	plugged into 100% power source	100%	As expected		Patrick Wang
5	accurate temperature sensing measurements	21.8 degrees C	21.8 degrees C	As expected		Patrick Wang
6	IC is on latest firmware update	IC flashed to latest firmware	IC updated	As expected		Patrick Wang
7	IC BAT+ charging voltage of 3.7V	3.7V into BAT+	3.7V into VC2	As expected		Patrick Wang
8	IC PACK+ input voltage 3.5 to 5.5V	CHG and DSG match Vin	CHG and DSG match Vin	they match indicating FETs open		Patrick Wang
9	IC PACK+ input voltage outside of range of 3.5 to 5.5V	CHG and DSG should be 0	CHG and DSG match Vin	to indicate FETs off		Patrick Wang
10	IC PACK+ Vin powers IC at same voltage	3V to 5V	3V to 5V	As expected		Patrick Wang

Table 1: Battery Gauge Validation

There are two main validation tests I had to run: 1) testing the I2C connection with the TI bqStudio software and 2) testing the input and output voltages against the expected measurements.

In regards to testing the I2C connection, I connected the SDA, SCL and GND pins on the board to the EV2400, which is a hardware evaluation tool used to communicate and configure TI-based battery management ICs. The EV2400 also has 10k pull-up resistors that “turn on” the data pins by increasing the voltage to 3.3V, which then connect to my laptop via a USB connection. From my laptop, the bqStudio application reads data from the IC. I was able to successfully connect my board to the bqStudio software and read some battery parameters as shown in the image below.

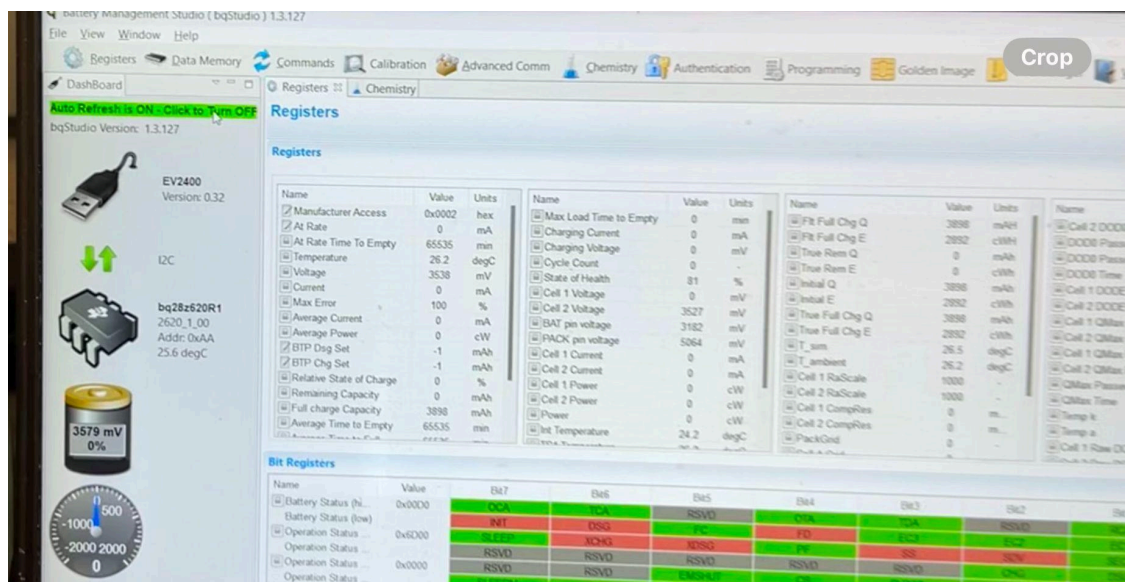


Figure 3: Battery Gauge in bqStudio

As can be seen on the left panel, the EV2400 and bqStudio are able to communicate with my bq28z620 board through a solid I2C communication. The battery charges at around 3.6 to 3.7V, which is expected, as the input voltage from the power supply is 3.7V.

The charging percent, however, shows 0% because the board is being connected to the DC power supply instead of the actual battery pack. This is because the battery holder I received from Amazon turned out to be a 3S1P cell configuration instead of a 1S3P configuration, which meant that the batteries were supplying 10V to my board which would fry the IC. I didn't have any other way to supply power to my board for the purposes of the demo as I didn't have a 1S1P battery holder I could use so the input would only be 3.7V, so I ended up using the power supply as the battery pack in my subsystem. Hence in this screenshot, my board is effectively charging the power supply with its own voltage, which has either resulted in the percentage showing 0% or 100%. This is normal, and when actually connected to a proper 1S3P configuration battery holder, the battery percentage will actually reflect the battery percentage of the batteries, not the power supply.

The four columns also show some information about charging parameters, including charging voltage, battery temperature, time until full charge, time until full discharge, as well as charge cycle count and other charging metrics.

In regards to testing input and output measurements against the expected measurements, the board is intended to receive current and voltage from the charger, and if those parameters are within certain ranges, the IC on my board would open the MOSFETs that effectively open the power path to allow for charging the batteries. If the inputs from the charger are outside the boundaries, the IC would close the power path and prevent charging. If the inputs are extremely high or extremely low, the board would enter "shutdown mode," and would send a fault signal to the MCU and immediately power off the entire system to protect against the fault.

The normal boundaries of operation are between 3.5 and 5.5V. This means that when given an input voltage of 5V, the CHG pin should show around 5V, as this would indicate that the MOSFETs have opened the power path and are allowing charging to the batteries. The inputs and outputs are shown here:

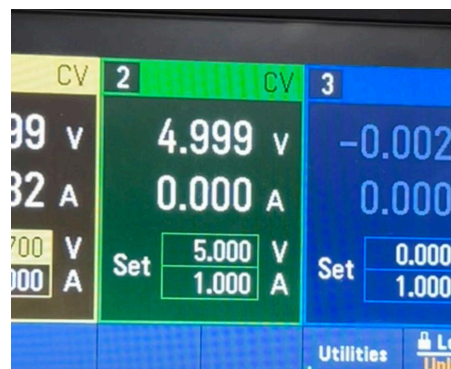


Figure 4: Charger Input with 4V



Figure 5: CHG pin showing MOSFET open, correct

In the photos above, given a 5V input, the CHG shows around 5V, which means that the MOSFETs are on. This is correct.

However, when I try to input a voltage out of range, the CHG pins should be off, but are instead still on.

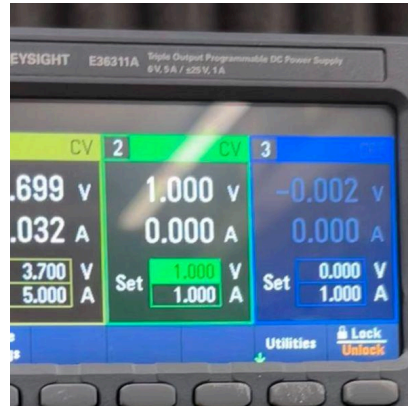


Figure 6: Charger Input with 1V

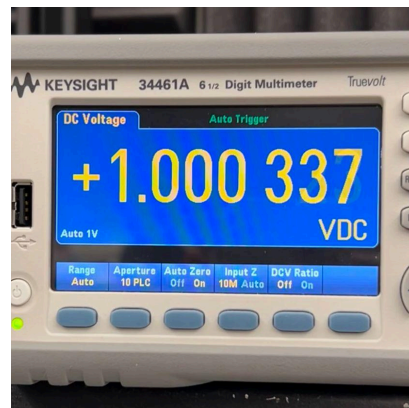


Figure 7: CHG pin showing MOSFET open, incorrect

As can be seen, with an input voltage of 1V which should trigger an undervoltage fault, the CHG pins still show 1V when it should be 0V, indicating the MOSFETs are still on.

The reason for this is because I had not yet configured the MOSFETs and initialized them properly in bqStudio. This is also mainly because I was only able to get the whole I2C communication with bqStudio to work just before the demonstration time.

2.4. Subsystem Conclusion

The battery gauge subsystem passed a huge portion of the validation tests, the biggest ones being confirming the I2C communication and getting the MOSFETs to open the power path to charge the batteries.

The only areas to work on are to 1) purchase the correct 1S3P battery holder so the battery gauge can start reading the battery pack parameters, and 2) to initialize the MOSFETs in bqStudio so the power paths will accurately open and close according to the input voltages by design. Both of these can be done very quickly, and with these changes, all of the validation tests will be successful.

3. Battery Charger - Cole Miller

3.1. Subsystem Introduction

The battery charger is a highly integrated switch mode buck converter. It can intake up to 18V and 3 amps input and output 3.5 to 4.8V to output. The charger can also be used to limit input voltage and current, measure the temperature of the battery, and also be used as a boost converter to take power from the battery to the input pin.

3.2. Subsystem Details

The schematic for the subsystem is provided below.

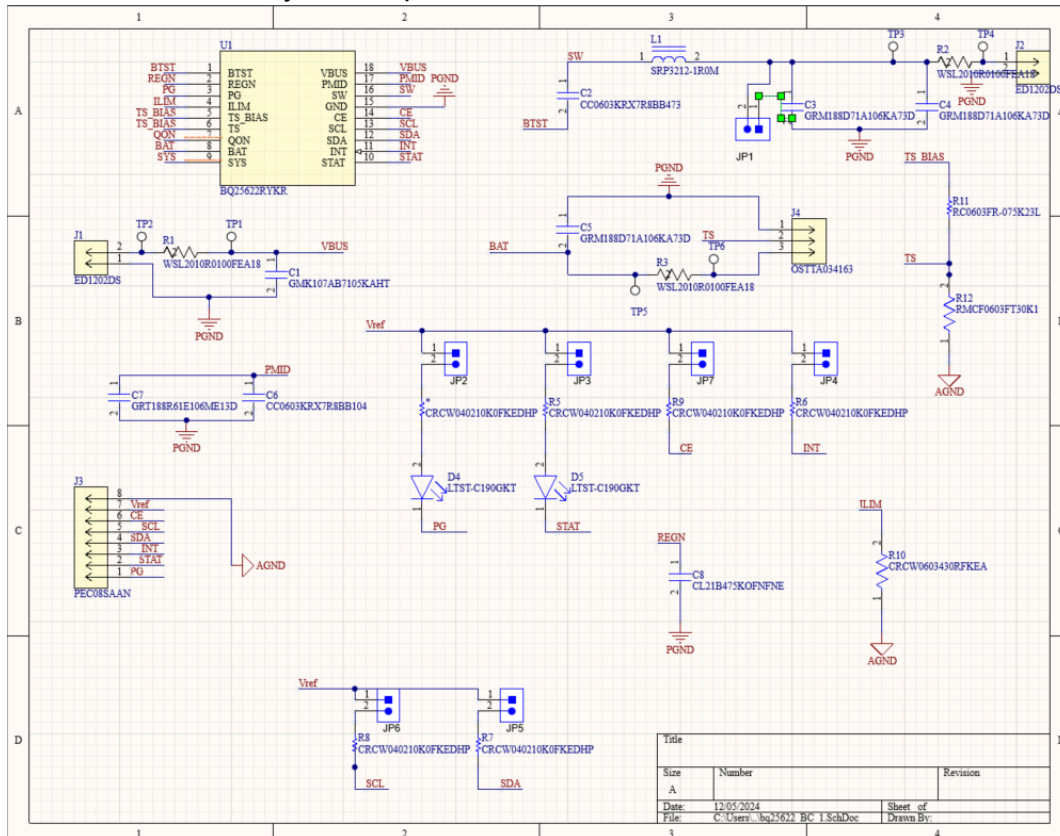


Figure 8: Charger Schematic

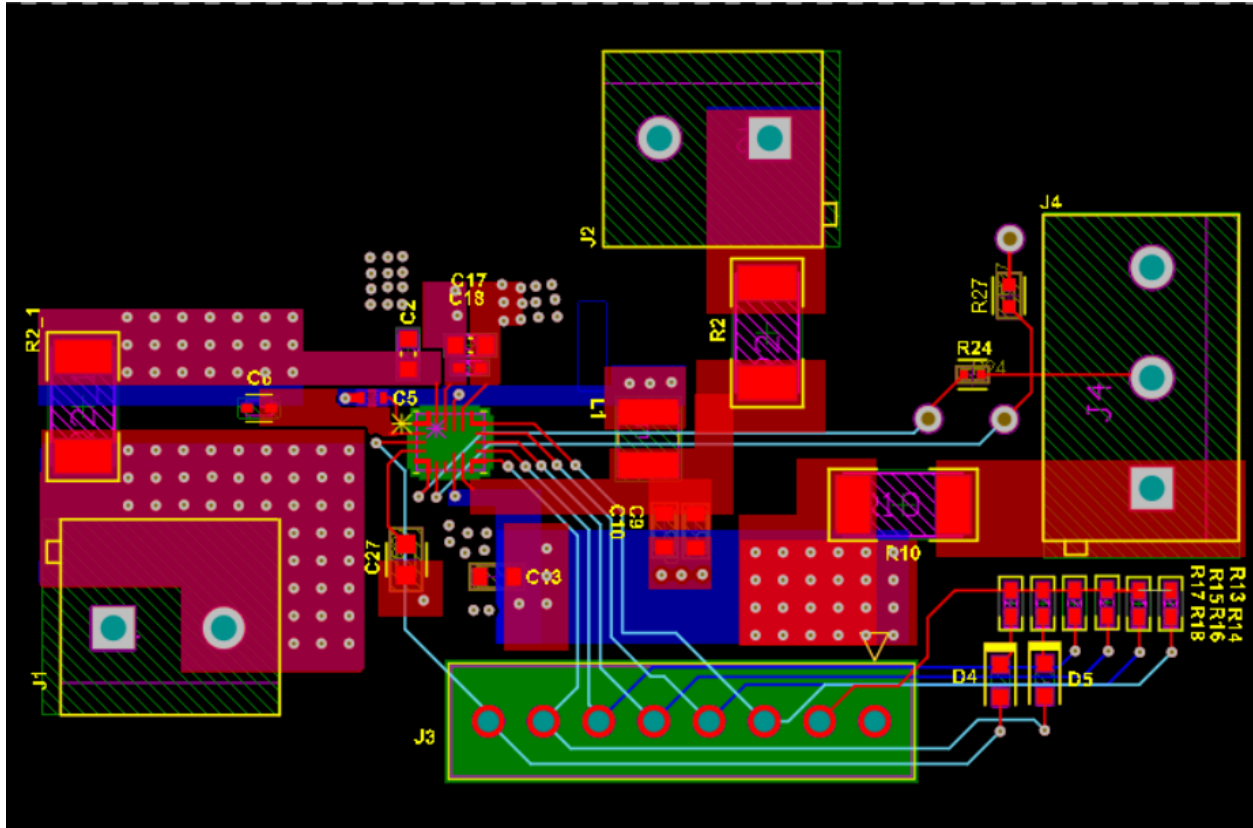


Figure 9: Charger Layout

The battery charger is built around a bq25622 IC. Utilizing I2C communication, a ADC, and a 12 bit register the charger can control the switching speed enabling it to control the exact current flowing through the charger. The charger acts as a way to set the voltage and current going into the battery to achieve the maximum power point transfer. The MCU would relay the needed current and voltage and the converter would set the parameters.

The charger also operates as a way to supply power to the USB-C port. The charger can enable OTG mode, this allows it to operate as a boost converter from the battery pin to the input pin and control voltage and current supplied to this pin. This allows the battery to supply 5v 1.5A to the input as that is the ICs limit.

In terms of layout it is a 4 layer board with the top and bottom layers mostly symmetrical, a power ground layer (removed to be more legible) a layer designated to analog signals. There are a lot of thermal vias and wide paths to dissipate heat as the board was 1 oz of copper.

The values were mostly chosen off of manufacturer recommendations. Capacitors were added to most pins to reduce switching noise. Then a 430 ohm resistor was added to the ILIM pin to hard limit the current to 3A.

3.3. Subsystem Validation

Number	Validation Test:	Input:	Expected Output:	Results	Status	Engineer
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1	Standard Operation	5V 3A on VBUS	4.2 V and 90% power efficiency	3.7V 2A on VBAT		Cole Miller
2	USB OTG Function	4.2V 1.79A on VBAT	5V 1.5A on VBUS	Unable to test		Cole Miller
3	ILIM Register Functionality	If input current >1A	output current >1A	Output limited to 2A since register can't be accessed		Cole Miller
4	VLIM Functionality	If Vin > 4.2V	Vo > 4.2V	Output limited to 3.7V		Cole Miller
5	PG Pin Functionality	Good input	Low signal	Low signal		Cole Miller
6	STAT Pin Functionality	Charging	Low signal	Low signal		Cole Miller
7	STAT Pin Functionality	Not Charging	High	High		Cole Miller
8	System Overvoltage Protection	Vin > Vsys_op	Vin > Vsys_op	Unable to test		Cole Miller
9	ILIM Hardware Functionality	Input current > 3A	Input current > 3A	Current limited to 3A		Cole Miller
10	IC valid I2C connection	IC connected to EV2400	IC communicates with BQStudio	doesn't communicate		Cole Miller
11	ILIM Pin Disable	Able to disable automatic current limit using register	Able to disable automatic current limit using register	Unable to control		Cole Miller
12	Switching Speed	1.5 MHz on SWS pin	1.5 MHz on SWS pin	1.5 MHz on SWS pin		Cole Miller

Table 2: Charger Validation

The charger in normal operation works exactly as expected. When voltage is applied to VBUS the charger bucks the voltage to the set level. This was validated with a test where the input was 5v 1.5683 A and the output was 3.699V and 1.9713 A as shown below. The frequency at the SWS pin was 1.5 MHz demonstrating that the circuit was on and charging.



Figure 10: Battery Charger Output

The only issue with the subsystem was a weak connection between the analog and power ground causing the digital pins to float. Connecting the two grounds as intended would cause the system voltage, the one used as reference voltage for the digital pins, to drop to 1.2V. However, once the grounds are separated and a separate power supply is used the voltage returns to normal and the circuit operates normally, besides the ability to communicate with the EV2400. By circumventing this problem the pins act normal, when 5V is added to STAT the charger turns off, and when voltage is added to the input PG turns to a low voltage.

3.4. Subsystem Conclusion

Overall the subsystem demonstrated its effectiveness as a charger. It could intake the voltage and output a highly efficient output to charge the battery. The only fixes that need to be made to the digital pin is to net tie the signal ground and power ground together, and use the voltage from the system pin as the Vref for the digital pins. This would resolve the problems with the digital pins and allow the control of the registers. Other improvements

needed is to try and reduce the side of the charger in order to make it more compatible to fit inside of the charger case.

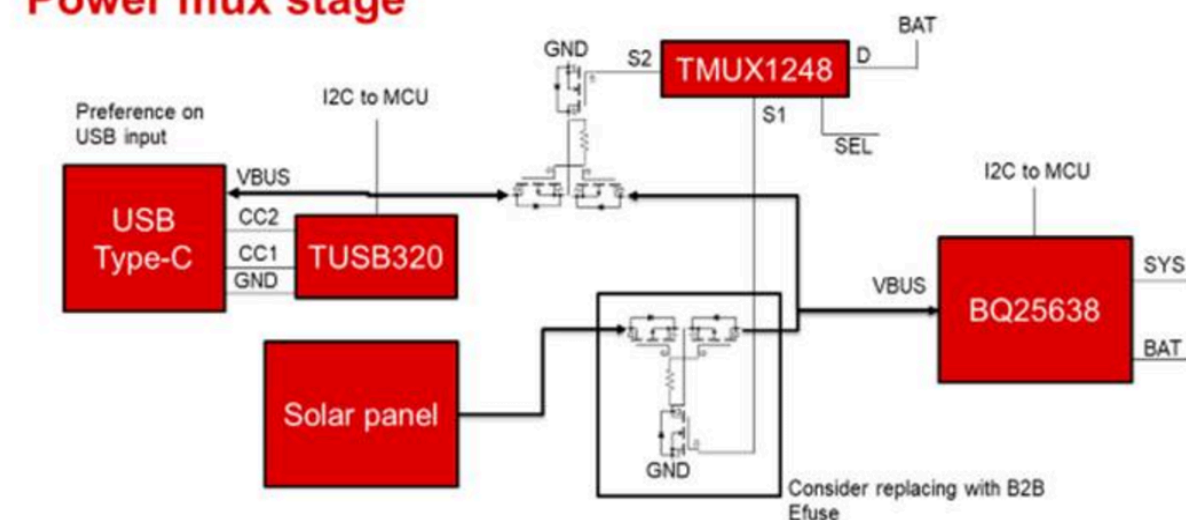
4. Power Mux - Camden Pomeroy

4.1. Subsystem Introduction

The power mux subsystem is designed to handle two power paths expected from the 15W solar panel and 15W USB type C input. This subsystem has two inputs and one output of VBUS that feeds into the battery charger subsystem. It is important that this subsystem can handle the expected 5V 3A inputs without shorting each other or overheating any subsystem. Finally the subsystem shall be able to handle bidirectional power flow from input to output and output to input.

4.2. Subsystem Details

Power mux stage



Reference: Achieve Bidirectional Control and Protection Through Back-to-Back Connected eFus

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TEXAS INSTRUMENTS

Figure 11: Block Diagram of Power MUX Subsystem

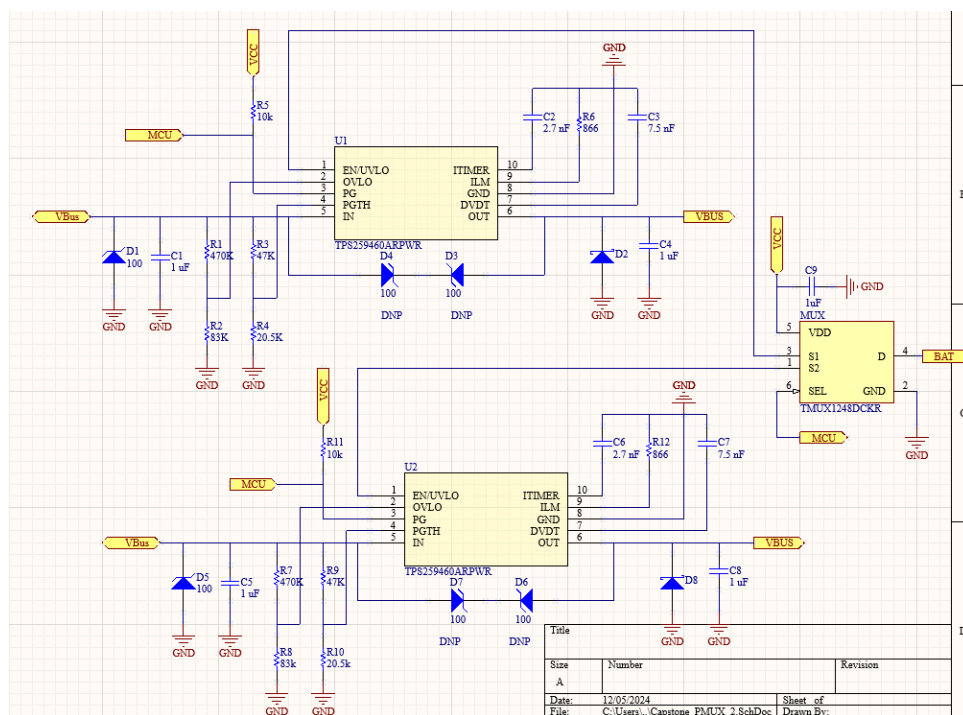


Figure 12: Schematic of Power MUX Subsystem

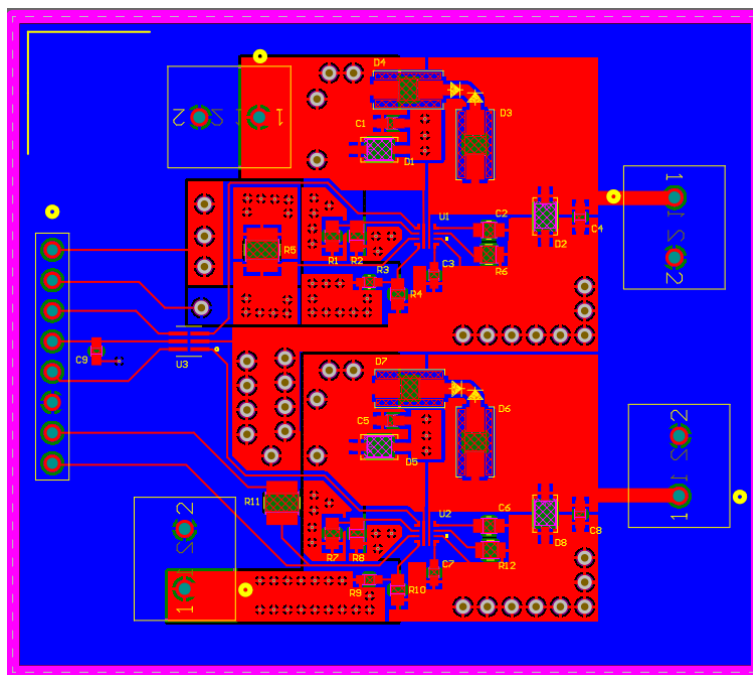


Figure 13: Layout of Power MUX Subsystem

The subsystem uses a TMUX1248 to control or enable each power path from input to output. Without the select pin high the Efuses pictured will be shut off. The Efuses chosen were the TPS25946 due to their ability to handle 2.7-23V and up to 5.5A. Other options included using other Efuses in a back to back configuration. These options were larger, more complicated, and more expensive than the TPS25946. Thus, the TPS25946 was

chosen. The TPS25946 offers undervoltage/overvoltage protection, current limiting and power good threshold features. The overvoltage was chosen to be 8V which is higher than the expected open circuit voltage from the solar panel. Current is limited to <2A during startup to limit any inrush current that could damage the subsystem. This ensures the safety and reliability of the entire subsystem. Additionally the power good threshold sets a reference voltage that when V_{in} is above, a digital output will be sent to the MCU indicating that the power on the given path is high enough to supply the battery. This can be used later for MCU logic and flow control.

The schematic design has two identical power paths in parallel with each other. Each path has one Efuse which has 10 pins. The EN/UVLO pin is enabled by the MCU by selecting a high or low voltage on the SEL pin of the MUX. The OVLO was set by choosing resistor values that would set the OVLO to 8V. This is higher than the highest expected voltage, thus the Efuse shall shut down operation. The PG pin is an output pin that indicates if VBUS is greater than the PGTH. The PGTH is the power good threshold pin which uses a simple voltage divider to set a reference voltage, currently 2 V. ITIMER, ILM, and DVDT pins are for the control and monitoring of the inrush current. The values of these pins were the recommended ones given in the application section of the datasheet for the TPS25946. Additionally, there are some extra zener and schottky diodes rated at about 14V to protect the circuit from voltage spikes. When a voltage spike occurs the diode will reverse activate and short to ground. Finally, recommended coupling capacitors rated from 0.1 μ F to 10 μ F were also added for even more power protection.

The layout design was created with large poly pours of copper to help dissipate the heat generated from these high current traces. Along with multiple stitching/thermal vias which help in larger current applications such as in the Power MUX subsystem. Capacitors were placed near pins when needed as in the case of V_{in} and V_{out} . As can be seen in **Figure 9** the layout for the USB C power path and solar Panel power path are nearly identical which makes design and validation much more efficient and simple.

4.3. Subsystem Validation

Below is a summary of the validation testing and results

Number	VALIDATION PLAN	Input:	Excpeted Output:	Results	Status
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	4.5V of Vout when input was 5V	
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected	
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is Low	As expected	
4	PP1 PG (Bad signal)	if Vin < 2V	PG signal is High	As expected	
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is High	As expected	
6	PP2 PG (Bad signal)	if Vin < 2V	PG signal is Low	As expected	
7	PP1 UVLO Operational	if Vin < UVLO	No output	As expected	
8	PP2 UVLO Operational	if Vin < UVLO	No output	As expected	
9	PP1 OVLO Operational	if Vin > OVLO	No output	As expected	
10	PP2 OVLO Operational	if Vin > OVLO	No output	As expected	
11	SEL PP1	SEL is high	PP1 is active and PP2 is not	Sel pin would not properly switch	
12	SEL PP2	SEL is low	PP2 is active and PP1 is not	Sel pin would not properly switch	
13	ILM durning startup	Inrush current of 4A	Limited current (<4 A) to ramp up	Could not get DC load to operate	
14	Seperate Power Paths	S1 active and both Vins 5V	Only PP1 Vout = 5V	As expected	
15	Seperate Power Paths	S2 active and both Vins 5V	Only PP2 Vout = 5V	As expected	
16	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected	

Table 3: Power Mux Validation

The validation testing results can be found above in **Table 3**. The performance of the board is largely as expected. When providing an input voltage of 5V I found that the output to be 5V. The PGTH, which sets the PG threshold, was 2V. When the Voltage fell below 2V the subsystem shut off and a high signal was sent to a pin. This pin represented the MCU in the case of total subsystem integration. When a voltage was applied to the output I found the input to also have the expected result. As far as the power path goes everything performed as expected. There was trouble with the TMUX1248 not properly selecting the right path to enable. The S1 pin was always high regardless of the voltage applied to the SEL pin. This is likely due to a short when testing. Additionally, one of the pads was lifted off of the board which held my last TMUX1248 so further testing could not be performed. Since each Efuse needs a voltage applied to their enable pins to operate and the TMUX1248 was supposed to supply that. I blue wired a connection directly to the enable pin so that further testing could occur. There were many issues with assembly that I had to troubleshoot with, especially with the QFN packages. The final issue that could not be fully tested was connecting a load to the output of my subsystem. I was only able to draw about 0.5A when my board is expected to handle 3A. Further testing is required to ensure that the board shall be able to withstand the full 5V 3A. Below are some pictures showing the various validation tests performed.

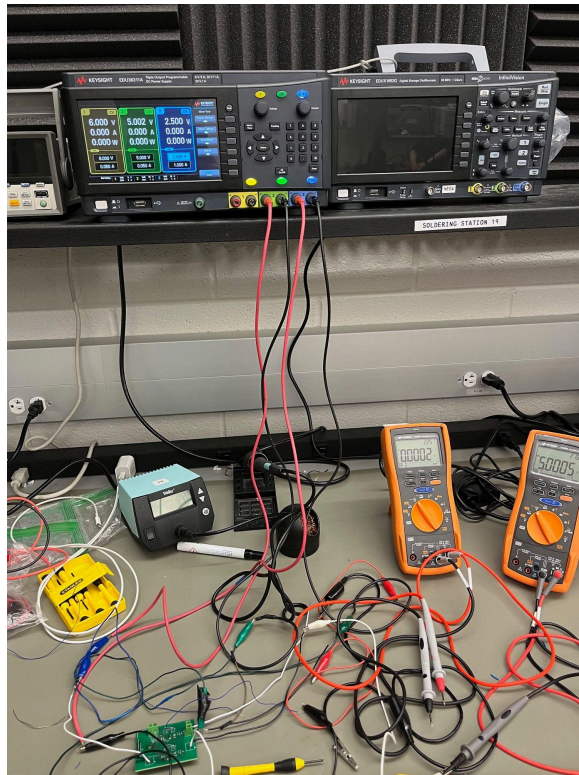


Figure 14: Validation of 5V input to 5V input (Enable high)

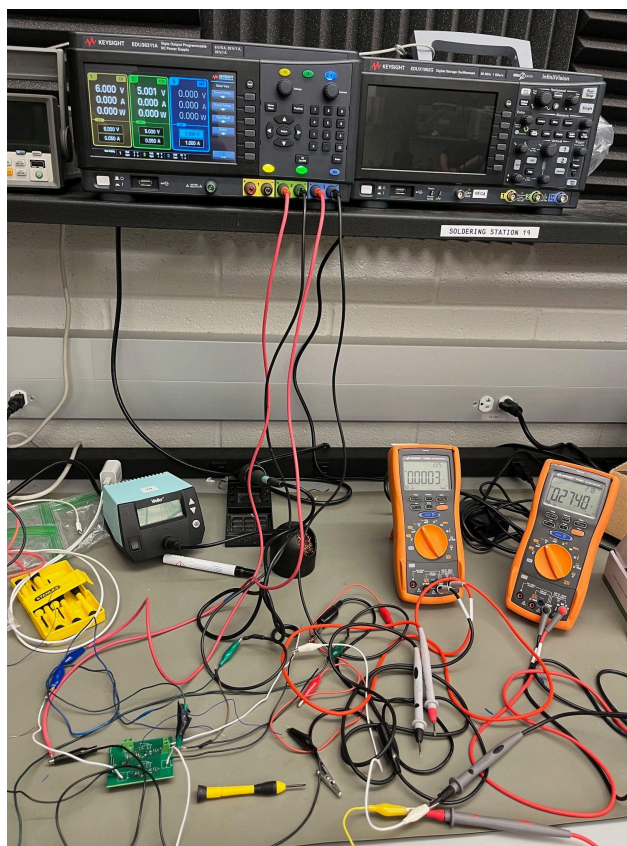


Figure 15: Validation of 5V input to 0V input (Enable low)

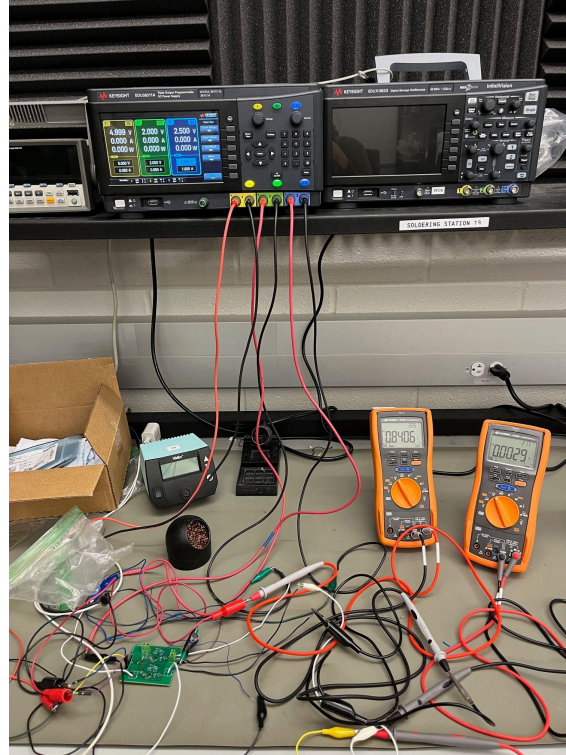


Figure 16: Validation of 5V input to PG high

4.4. Subsystem Conclusion

The subsystem design was shown to handle two inputs one output with bidirectional power flow. There were some major assembly issues with the QFN packages that caused the inoperability of the switching of power paths. All safety protection and reliability features were validated to perform as expected. The enable pins on the TPS25946 perform as expected, however further validation is needed to be done on the switching of power paths after more TMUX1248 parts are delivered in order to fully validate the subsystem.

5. MCU - Jaein Cha

5.1. Subsystem Introduction

The Microcontroller Unit, or MCU for short, is a core brain unit of the battery pack system for decision making and control. Using I2C as a communication method, the MCU integrates all subsystems including the battery charger, power MUX, and battery gauge, processes incoming information from such subsystems to monitor current status of the battery pack, and operate protocols with other subsystems to eventually provide stable power and information such as charge status of the batteries to users. For its functionality and validation, it has a SOP display and TUSB320 USB-C port controller as part of the subsystem.

5.2. Subsystem Details

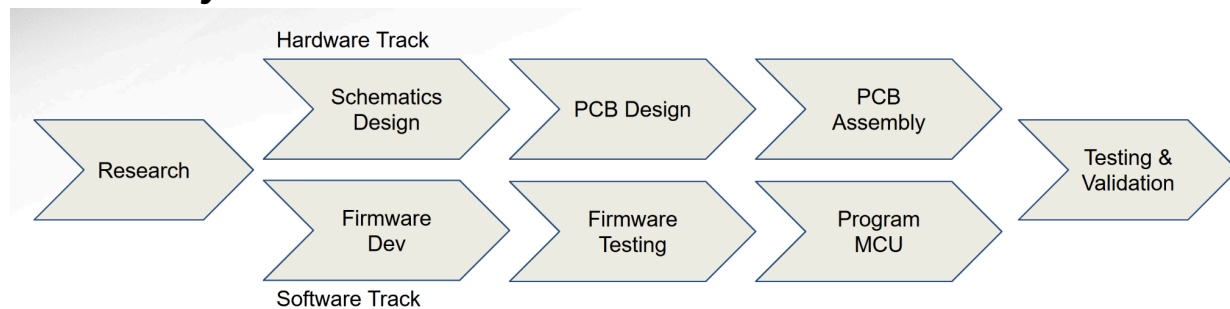


Figure 17: MCU subsystem development milestones

By nature of the MCU, a unit in charge of decision making of the entire system and has a wide spectrum of possible applications, it requires custom firmware to be developed unlike other subsystems. Due to such requirements, the MCU subsystem had a two-track development process, having tasks for software (firmware) development in addition to the hardware (custom circuit board) development. Figure 17 visually shows the milestones of the MCU subsystem development process.

Briefly, the MCU's main task can be divided into two: initiating I2C communication as a I2C controller and the Maximum Power Point Tracking (MPPT) management. Because implementation of MPPT requires close collaboration and connection between the battery charger, battery gauge, and the MCU, the MPPT implementation is planned to be done in ECEN 404. Instead, implementation of the custom PCB and the I2C firmware development was the main focus of ECEN 403. Specifically, the goal of ECEN 403 for the microcontroller unit subsystem was to build a functional, successfully programmed MCU custom PCB that can manage correct bi-directional communication between different ICs and users as a I2C controller and also through direct GPIO controls.

The following Table 4 and 5 shows comparisons among the hardwares and platforms initially considered to implement the required functionality of the MCU - integration of the subsystems via communication protocol and informing users regarding the status of MCU and the battery pack in the near future. The tables were made during the research phase of the project, organizing pros and cons of each considered option.

Type of Hardware	Parts	Pros	Cons
MCU	MSPM0L1306	Less power usage	Less functionality
	MSPM0G3507 - VSSOP (DGS) Package	More functionality (such as math accelerator) Easier to solder than packages with more pins	More power consumption (compared to MSPM0L) Less GPIO
	MSPM0G3507- RHB Package	More functionality (such as math accelerator) More GPIO	More power consumption (compared to MSPM0L) Harder to solder
SOC Display	LCD Display	More detailed information can be delivered	Takes more power Vulnerable to damage
	5 LEDs	Robust to damage, long-lasting Takes less power Easier to program for it	Limitation in details of information that can be delivered

Table 4: MCU subsystem - hardware comparison chart

Type of Platforms	Parts	Pros	Cons
Communication Protocol	SPI	Faster communication	Need more pins
	I2C	Multiple device connection granted by one set of pins Less pin usage Simpler to implement	Slower communication
Programming Language	Embedded C	Faster execution speed Lots of resources/tutorials available	Not natively object oriented, yet still can be achieved
	Embedded C++	Natively Object-Oriented Programming language	Less examples/resources available Heavier to run, slower

Table 5: MCU subsystem - platform comparison chart

In table 5, on the other hand, software platforms such as communication protocols and programming languages to be used are compared. For communication protocol, because the chosen package of the MCU has fewer pins, I2C was chosen to conserve the amount of used pins as I2C requires only two pins to integrate all the subsystems. Also, embedded C was chosen as a programming language over C++ since it is faster than C++ and widely used and therefore there are more examples available on the internet.

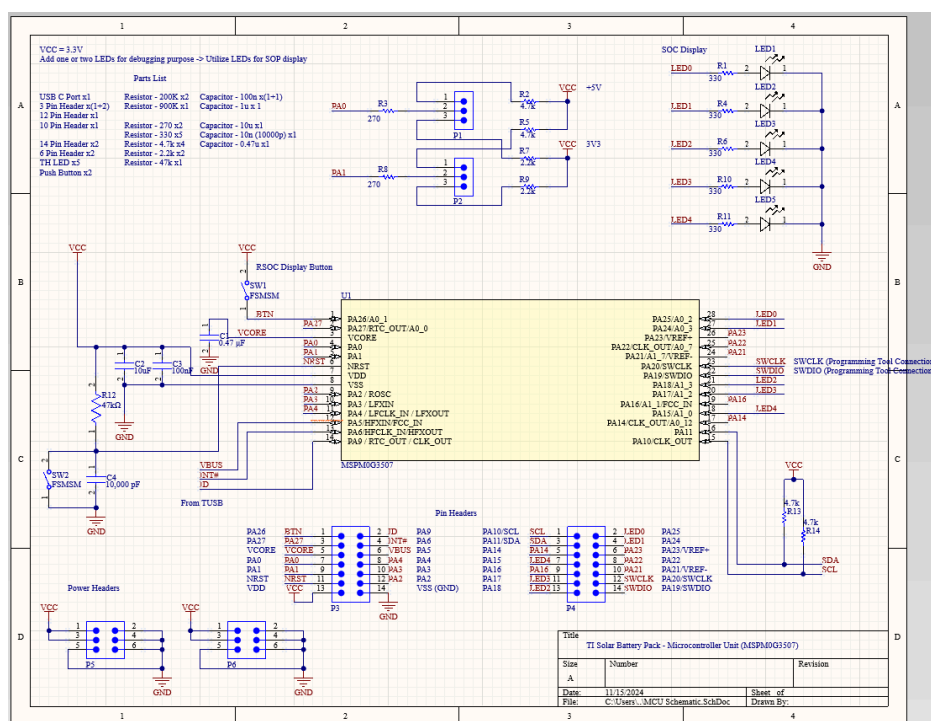


Figure 18: MCU Schematic I - MSPM0G3507

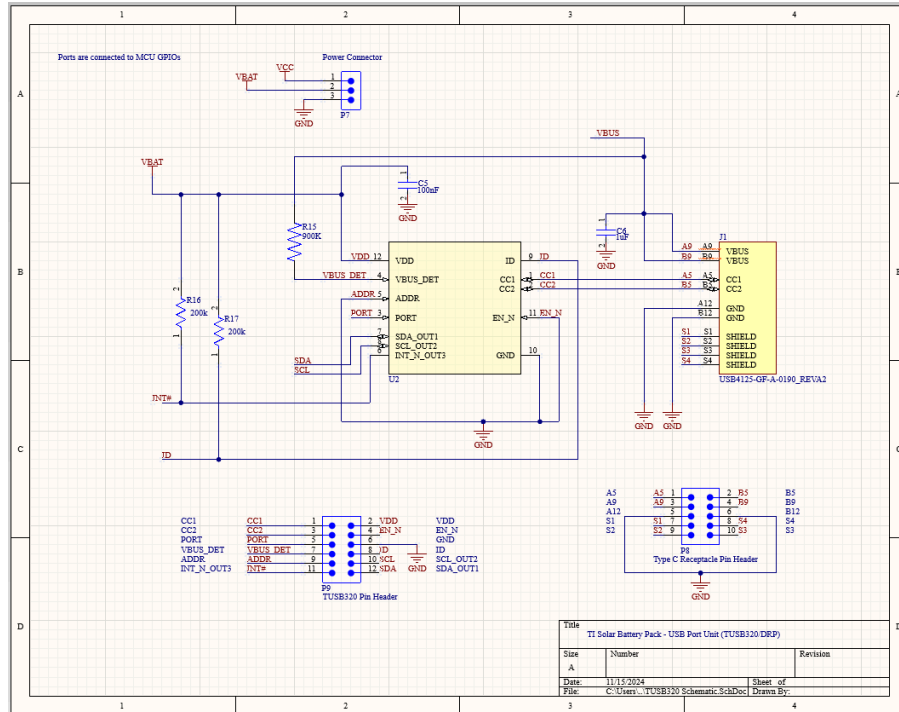


Figure 19: MCU Schematic II - TUSB320

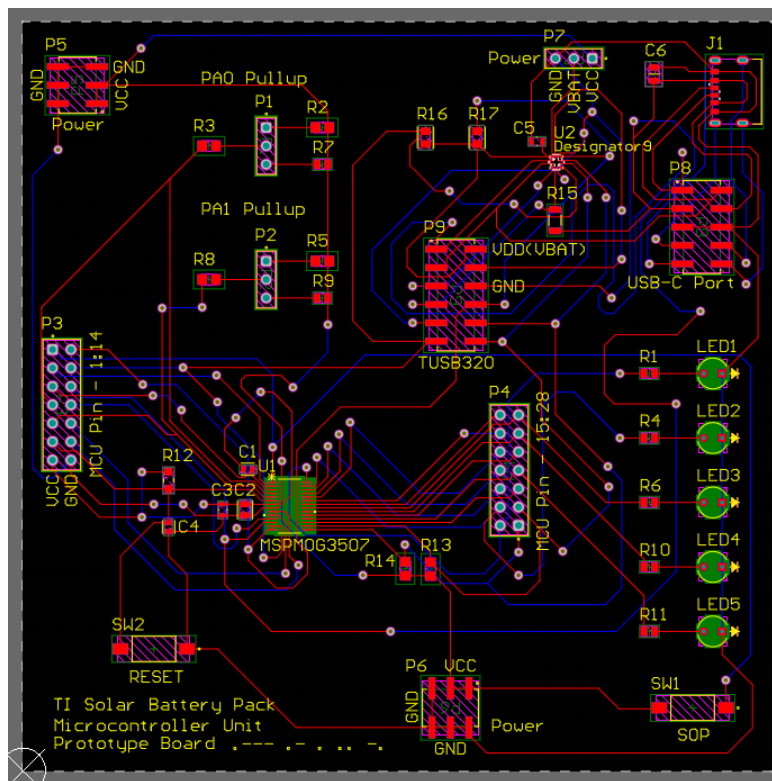


Figure 20: MCU Layout

After deciding all the major parts in the research phase, I moved onto firmware development and schematic design. During this process, the sponsor suggested having a USB C port

controller IC, which is TUSB320, on the PCB to enable easier future testing of working I2C protocol on board. Based on the suggestion, two schematics for the subsystem were made - Figure 18 for MSPM0G3507, and Figure 19 for TUSB320. Generally, both schematics are designed based on the datasheets, with additional pin headers for power connections and debugging purposes. TUSB320 schematic design heavily referred to the example port controller board design for dual-role power (DRP) port attached in its datasheet, including the necessary resistors and capacitors and their values. The schematic was designed to implement a DRP port since the battery pack will be charged and charge other devices through the same USB C port.

However, due to high flexibility in the MCU's application, there were no reference examples found from the datasheet for the MCU schematics. Instead, to design the schematic, both the datasheet of MSPM0G3507 and its development board, TI MSPM0G3507 launchpad's schematic, was referenced as well as some general design guidelines for custom PCB designs available on the internet. In the MSPM0G3507 schematic (Figure 18), LEDs and a SOP button were located along the MCU and header pins to meet its expected functionality: user input device (the button) and the output device (LEDs as the SOP display). For the values of pull-up resistors for I2C pins and a few GPIOs, decoupling capacitors, and capacitor-resistor set for the NRST(reset') pin followed suggested values found in the data sheet. Moreover, design of disconnectable pullup resistors for PA0 and PA1 referred to the development board's design for the future flexibility in connecting different subsystems taking unforeseen circumstances or requirements of GPIOs into account. The value for the resistors that connect the LEDs and GPIOs are set to a very small value, considering the small current flowing from the MCU. VCC is set to 3.3V following the suggested value in the datasheet.

After the schematics design, PCB design was done; the final layout can be found as Figure 20 above. PCB routing was manually done on Altium Designer, ensuring that the design rules are followed and required clearances are kept. Also, considering the board is two-layered, having segmented ground planes were avoided in order to reduce impedance caused by having unintended loops made of the segmented ground planes.

The firmware for I2C communication protocol was implemented on TI CCS Theia IDE following available examples online, but the code is only dummy-tested, due to a connectivity issue on board. The problem will be described with more details in the next subsection.

5.3. Subsystem Validation

Number	Validation Test:	Input:	Expected Output:	Results	Status	Engineer
1	Continuity test - before soldering	Continuity test using DMM	Beep (connected) for same nets, no sound otherwise	As expected		Jaein Cha
2	Continuity test - after soldering	Continuity test	Beep (connected) for same nets, no sound otherwise	As expected		Jaein Cha

		using DMM				
3	Voltage/current test (MCU)	3.3V to the power pin (VCC)	3.3V from the source GPIO pins	As expected (from SCL pin)		Jaein Cha
4	Voltage/current test (TUSB)	3.3V to the power pin (VCC)	3.3V from the source GPIO pins	As expected (from ID pin)		Jaein Cha
5	I2C - voltage level	3.3V to the power pin (VCC)	Same voltage of 3.3V for both ends of I2C bus	As expected		Jaein Cha
6	I2C - initialization	3.3V Power supply to the power pin (VCC)	Read TUSB I2C address	TUSB unresponsive		Jaein Cha
7	I2C - receive (DFP connected)	DFP plug is connected to the port	Read TUSB I2C address and value, blinks LEDs corresponding to the code for "DFP inserted"	TUSB unresponsive		Jaein Cha
8	I2C - receive (UFP connected)	UFP plug is connected to the port	Read TUSB I2C address and value, blinks LEDs corresponding to the code for "UFP inserted"	TUSB unresponsive		Jaein Cha
9	I2C - receive (DRP connected)	DRP plug is connected to the port	Read TUSB I2C address and value, blinks LEDs corresponding to the code for "UFP inserted"	TUSB unresponsive		Jaein Cha
10	Reset	Reset button pressed	MCU resets and stop working while	Yes on LP, not on		Jaein Cha

			the button is pressed	custom board		
11	SOP Display	SOP button	Lights up all the LEDs when SOP button is pressed	Yes on LP, not on custom board		Jaein Cha

Table 6: MCU Subsystem Validation

Right before and after assembly of the custom PCB, validation testing was done as shown in Table 6. Table 6 contains planned validation and testing with expected input and results. During validation, continuity tests of the board before and after assembly were done to check if the manufactured board is defective and there were any problems during soldering. These tests are basic but important, as correct functionality expects nets being continued as expected, without being shorted with other nets. Then, voltages of certain spots such as VCC, GND, source GPIO pins, I2C bus, on the board were tested to check if the power was provided correctly. Below figures show some of the outputs of the validation - especially voltage tests.

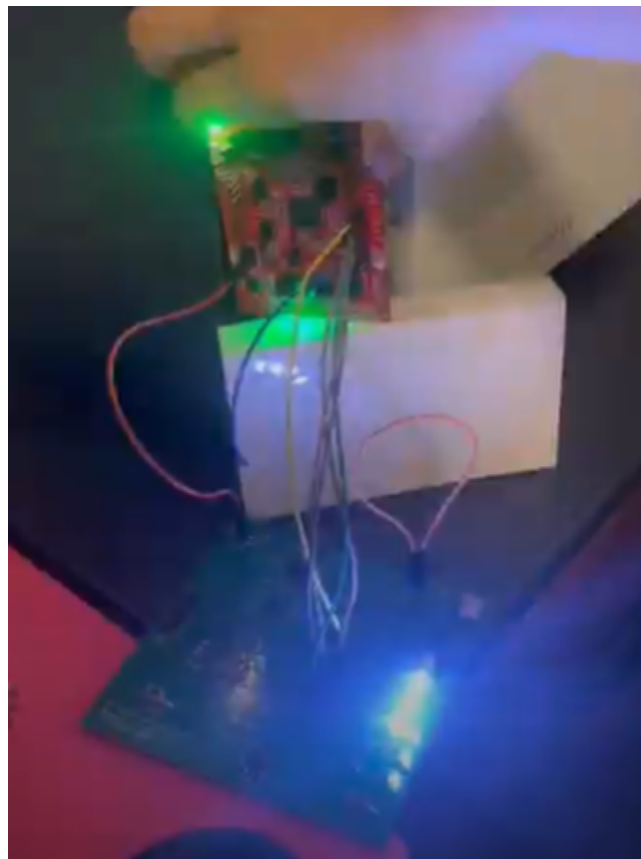


Figure 21: MCU SOP Display - Button Response Test

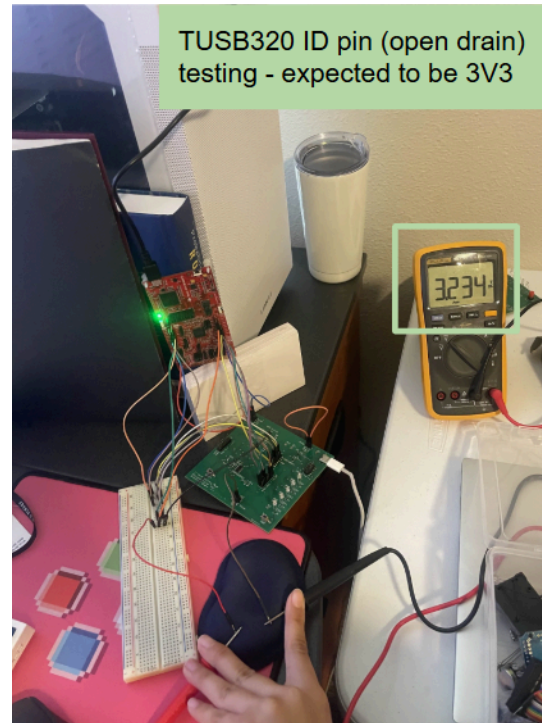


Figure 22: MCU Voltage Test I

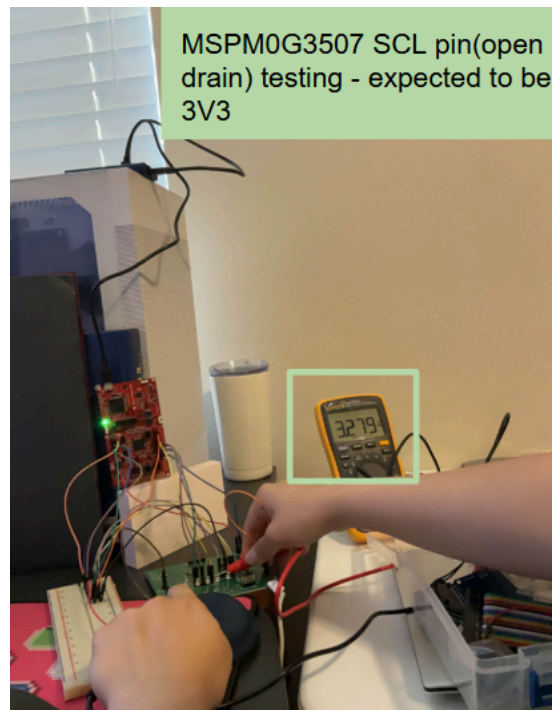


Figure 23: MCU Voltage Test II

All the continuity tests and voltage tests passed without any problem, however, tests involving I2C which requires connections of the ICs had a problem getting expected results. This implies either a connection problem between pads and the ICs or IC failure due to long

exposure to high temperatures from the soldering process involving the usage of a heat gun. Since the soldering process was done using a microscope and it seems that there is a visible amount of solder, and also because there were no shorts detected between IC pins, it would be more likely that the problem is caused by the second reason.

Due to the connectivity problem, the MCU on the board couldn't be programmed and tested per validation plan. Having the same problem, TUSB320 was also not responsive from the I2C communication control initiated by the development board, MSPM0G3507 Launchpad. Because of this, I2C firmware wasn't able to be tested on board. I focused on debugging the hardware before the demo so I couldn't think of testing the firmware on the EVMs we have (such as the battery gauge), but during the meeting with the sponsor after demo, he suggested that testing of I2C using other EVMs for different subsystems.

To fix the problems found during validation and testing, I plan to re-solder new ICs to resolve any problems caused by the soldering process. Also, in parallel, firmware testing will be done on actual boards, not by dummy signals given by myself, using other EVMs we have for battery gauge and charger.

5.4. Subsystem Conclusion

Based on the given scope and purpose of the system by the sponsor, the microcontroller unit custom printed circuit boards and the firmware for subsystems communication were designed to achieve the MCU's ability to control all subsystems and manage information processing among the battery pack system, and then tested with the validation plans. Due to unfamiliarity in designing embedded systems and custom PCB boards, there was a delay in the development progress, and a major connectivity issue happened during assembly. Currently, the code is working but only tested with dummy signals generated by myself, however, to prevent any delay of the progress in ECEN 404, the I2C firmware will be tested with battery gauge's EVM and the board will be debugged over the winter break. The goal over the winter break is to debug the hardware and fully validate functional PCB before next semester starts.

6. Solar Panel - Cole Miller

6.1. Subsystem Introduction

The goal for the solar panel was a compact 15 watt solar option. This would allow fast charging while not impeding travel for the user.

6.2. Subsystem Details

We used a FlexSolar 15 watt solar panel as our solution. This was a highly compact at 8.8" by 7.8" by 0.4" solar panel that only weighs 1 pound. It was quoted to provide 5V 3A charge while also being USB compatible to allow easier testing and possible integration using a standard USB port.

6.3. Subsystem Validation

Number	Validation Test:	Input:	Expected Output:	Results	Status	Engineer
1	Solar Panel VOC	Solar panel Voc < 18V	Solar panel Voc < 18V	As expected		Cole Miller
12	Solar Panel Power	regular usage in sunny day	output 15W	As expected		Cole Miller

Table 7: Solar Panel Validation

In terms of necessary validation the solar panel provided 7.37 volts and provided 2.115 amps meeting the necessary 15 watts. The current was calculated as the battery charged 7% of a 2517 mAh battery in 5 minutes. The solar panel also outputs .3256 in cloudy conditions.



Figure 24: Reference Documents

6.4. Subsystem Conclusion

The solar panel acquired fits the criteria for the project and also provides the necessary power. The only thing left for next semester is to measure the lumens the solar panel is under to characterize it based on lumen then just based on weather. This would allow a more accurate characterization and better predict charge efficiency in adverse conditions.

TI Solar Charged Battery Pack

Jaein Cha
Cole Miller
Camden Pomeroy
Patrick Wang

SYSTEM REPORT

REVISION – 2
28 April 2025

SYSTEM REPORT
FOR
TI Solar Powered Battery Pack

TEAM 64

APPROVED BY:

Patrick Wang Date

Prof. Kalafatis Date

Ali Alinezi Date

Change Record

Rev.	Date	Originator	Approvals	Description
1	4/18/2025	TI Solar Powered Battery Pack	Patrick Wang	Draft Release
2	4/28/2025	TI Solar Powered Battery Pack	Patrick Wang	Revision 1

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1 Subsystem Changes

1.1 Power Mux - Camden Pomeroy

To fully meet the requirements of the power mux system specified in the FSR and by the sponsor some changes were required to be made to the subsystem. The TMUX1248 would frequently short itself and cause two loads to connect to the battery charger at once. To compensate for this I increased the trace widths on the pcb in and out of the TMUX1248. This solved the problem.

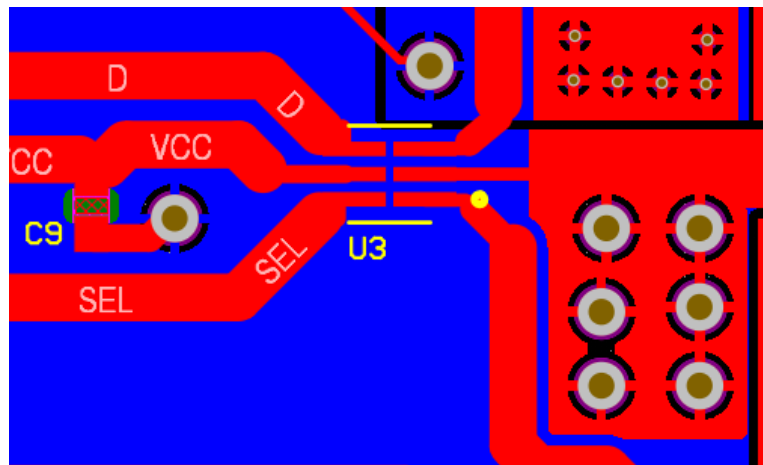


Figure 1: PCB view of TMUX 1248

Previously, the power mux would only allow power flow in one direction, from in to out. Many options were examined including the reconfiguration of the entire subsystem. However, after careful consideration and rereading of the datasheet we noticed that adding a diode and resistor in series from out to in would bias the in TPS25496 to enable a low impedance path and allow bidirectional current.

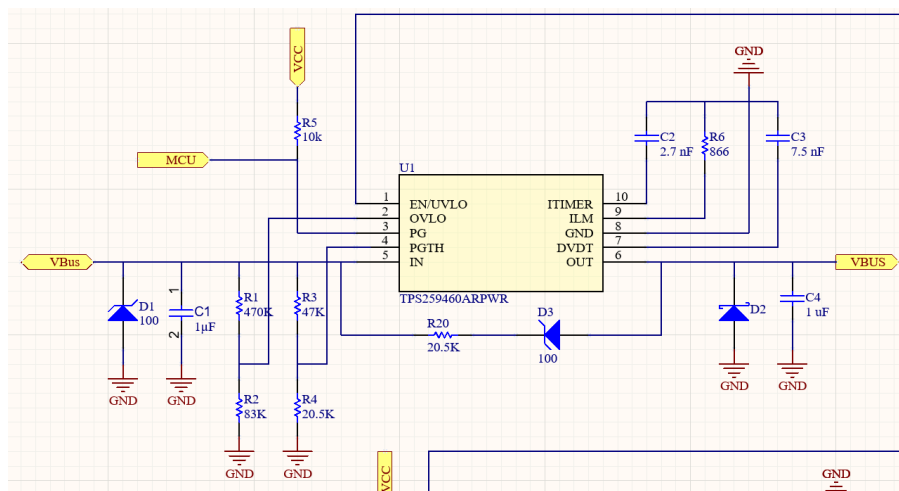


Figure 2: Schematic view of Changes made to Power Mux Subsystem

Finally, some general design changes were made to improve reliability and the efficiency of my subsystem. Originally, I was getting a voltage drop on my subsystem likely due to overheating which caused a loss of some power. Adding some thicker polygon pours of copper onto the pcb and allowing for thicker traces reduced that voltage drop significantly. Additionally, the solar panel would frequently trigger the overvoltage and undervoltage conditions of the eFuses. To counteract this I changed the overvoltage and undervoltage thresholds of my system. This was done by changing the values of R1, R2, R3, and R4. The values became 2V and 8.5V, so the solar panel would stay within the nominal range of the subsystem. Some minor fixes to the schematic were also implemented such as increasing the capacitor sizes on VBUS to better smooth out the input and output voltages.

1.2 Battery Charger - Cole Miller

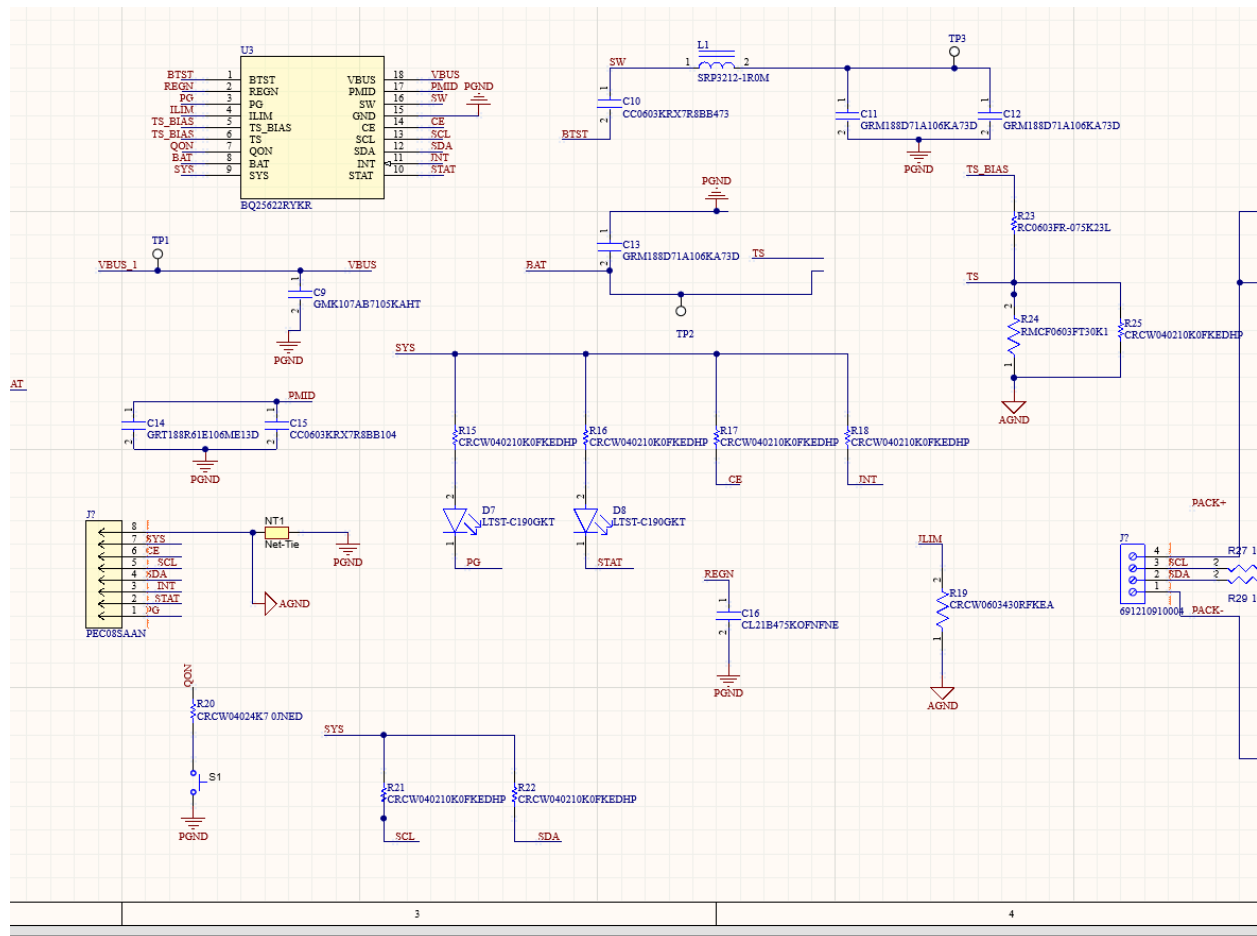
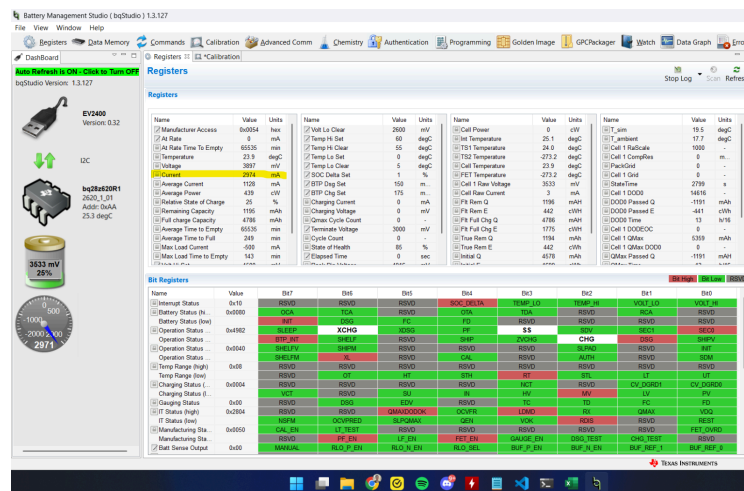


Figure 3: Schematic view of the Battery Charger

Previously, the battery charger lacked a net-tie between the signal and ground plane. The addition of that helped the charger communicate and properly measure voltages utilizing its ADC. The only other additions was a reset button in case it was ever needed and the removal of the current sense resistors that were used for testing the original board. No components were changed and the PCB schematic remained relatively unchanged.

1.3 Battery Gauge - Patrick Wang

In regards to changes in the battery gauge subsystem, there were some challenges and differences I had to account for. One main challenge was calibration of the gauge to be able to read the appropriate charging voltage and current, as sometimes it would read values that were slightly off by a couple hundred mV. This would cause the gauge to trigger overvoltage or undervoltage faults in certain conditions. I eventually found a solution in programming to the correct bqz file associated with the gauge, and if that didn't work, I would just swap the IC itself. I ran through many ICs before figuring out that a lot of the issues came from lack of proper firmware, or out-of-date firmware. The figures below show the now calibrated gauge being able to read the appropriate charging voltage of the batteries, along with the ability to appropriately enable and disable FETs to open the power paths when given voltages that are both within and outside of the set 2.5V to 4.3V operating threshold.



In Figure 4, when given an input of 4V3A, which is within the operating threshold set at 2.4 to 4.3V, the gauge can be seen charging at around 3.95V, which is accurate. The highlighted line shows the correct charging current at 3A, indicating that given a 4V3A input, the gauge registers the correct 3A input.

Figure 4: Input 4V3A (charging)

In Figure 5, when given an input of 2V3A, which is outside the operating threshold, the gauge can be seen not charging anymore, as the voltage drops to 3.5V (which is simply just the voltage of the battery as it cannot charge), and the highlighted area shows the current is near 0, which indicates that the battery gauge is working as intended.

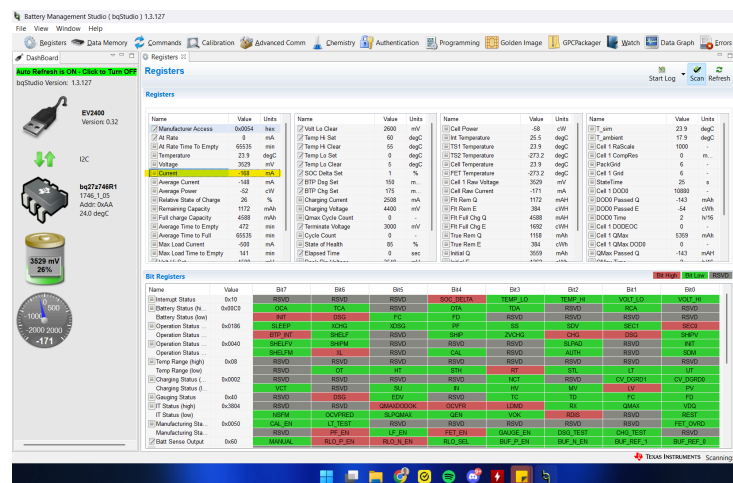


Figure 5: Input 2V3A (not charging)

To minimize the difference in firmware implementation in the launchpad and the on-board MCU, the on-board MCU package has been changed to the same one as on the launchpad. Therefore, the on-board MCU package is changed from the previous package (VSSOP (DGS), 28 pins) to the current package (LQFP (PM), 64 pins), and the layout has been redesigned accordingly, which also took the PCB integration of all the subsystems into consideration. Figure 4 is the updated schematic for the MCU, using LQFP (PM), 64 pin package.

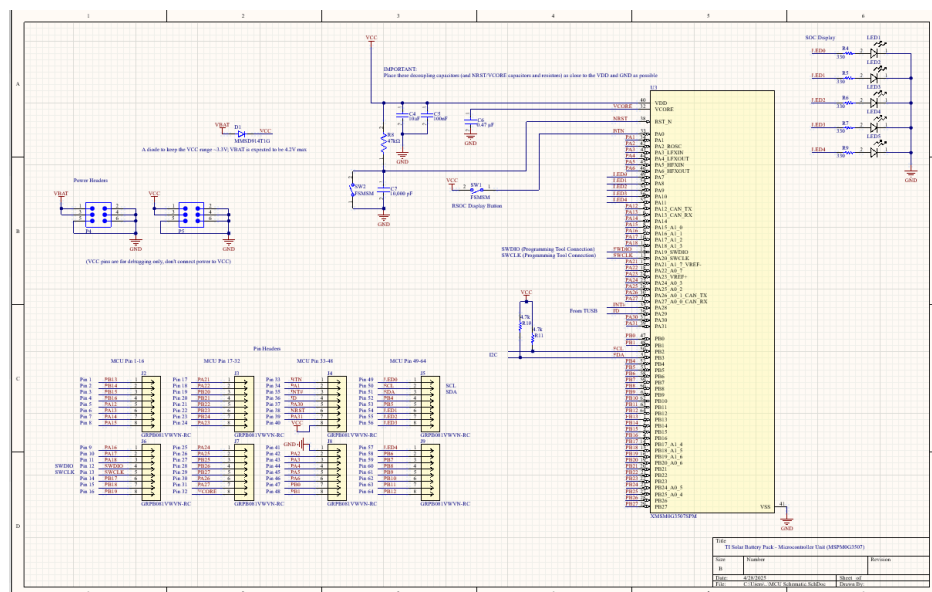


Figure 6: Schematic view of the Microcontroller

Additionally, for the correct implementation of DRP configuration with TUSB320, the USB-C port controller IC, a NMOSFET has been added and connected to a bulk capacitor and VBUS, with its gate connected to one of the MCU's GPIO, as shown in Figure 5, the updated schematic of the TUSB320. Moreover, to keep the MCU's supply voltage (VCC) within the operating range (around 3.3V), a diode has been placed between the VBAT net and the VCC net, with the purpose of regulating voltage of VCC to have 0.3-0.7V lower than VBAT (voltage of the battery, expected to be 3.7V with maximum value of 4.2V).

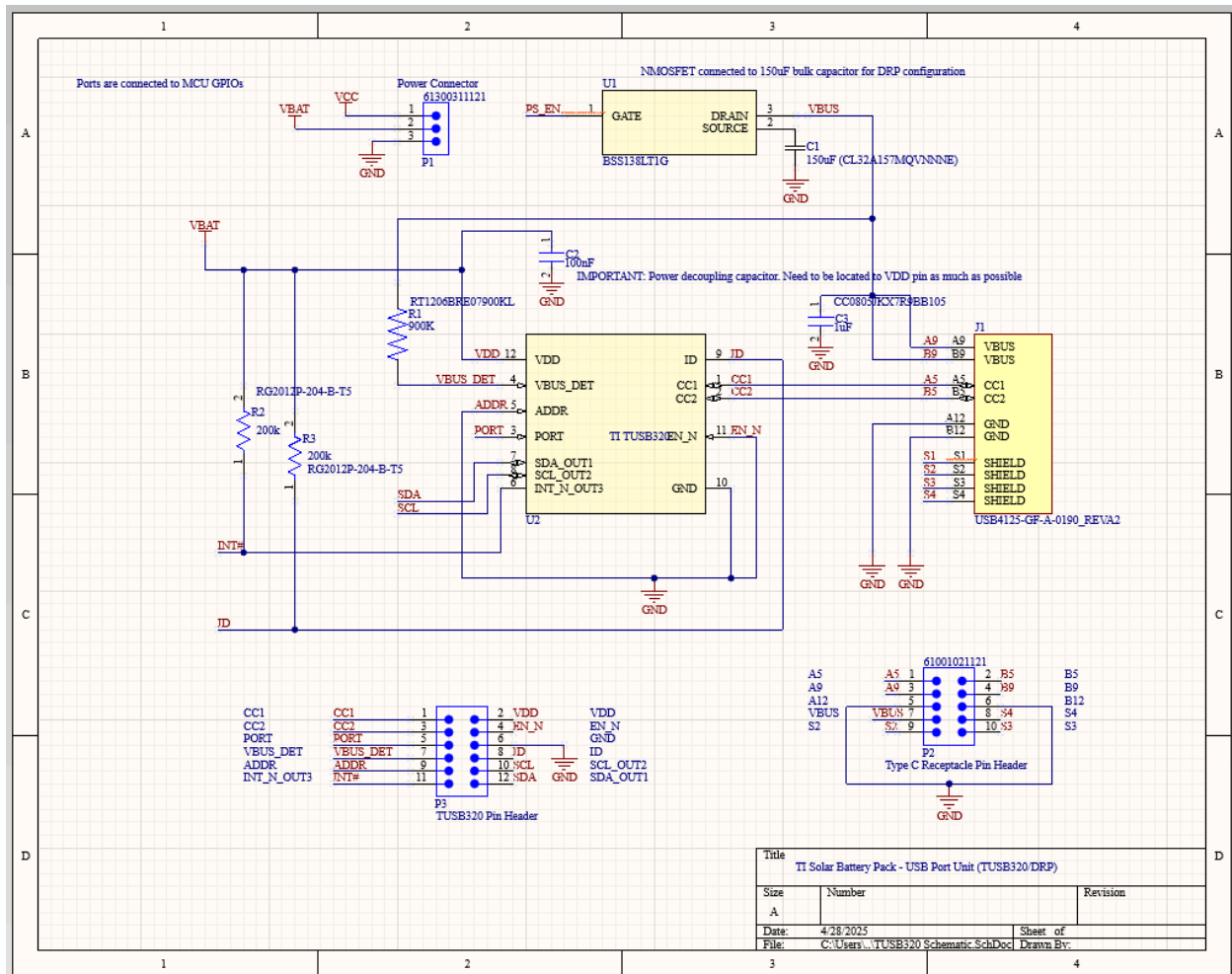


Figure 7: Schematic view of the USB-C Port Controller

In terms of software, the firmware consists of four big parts: control logic, SOC display management, safety guarantee feature, and Maximum Power Point Tracking (MPPT), which have been fully implemented and integrated. Figure 6, the flowchart of the MCU firmware, portrays the control and safety logic of the implemented software.

MCU Firmware Flow Chart

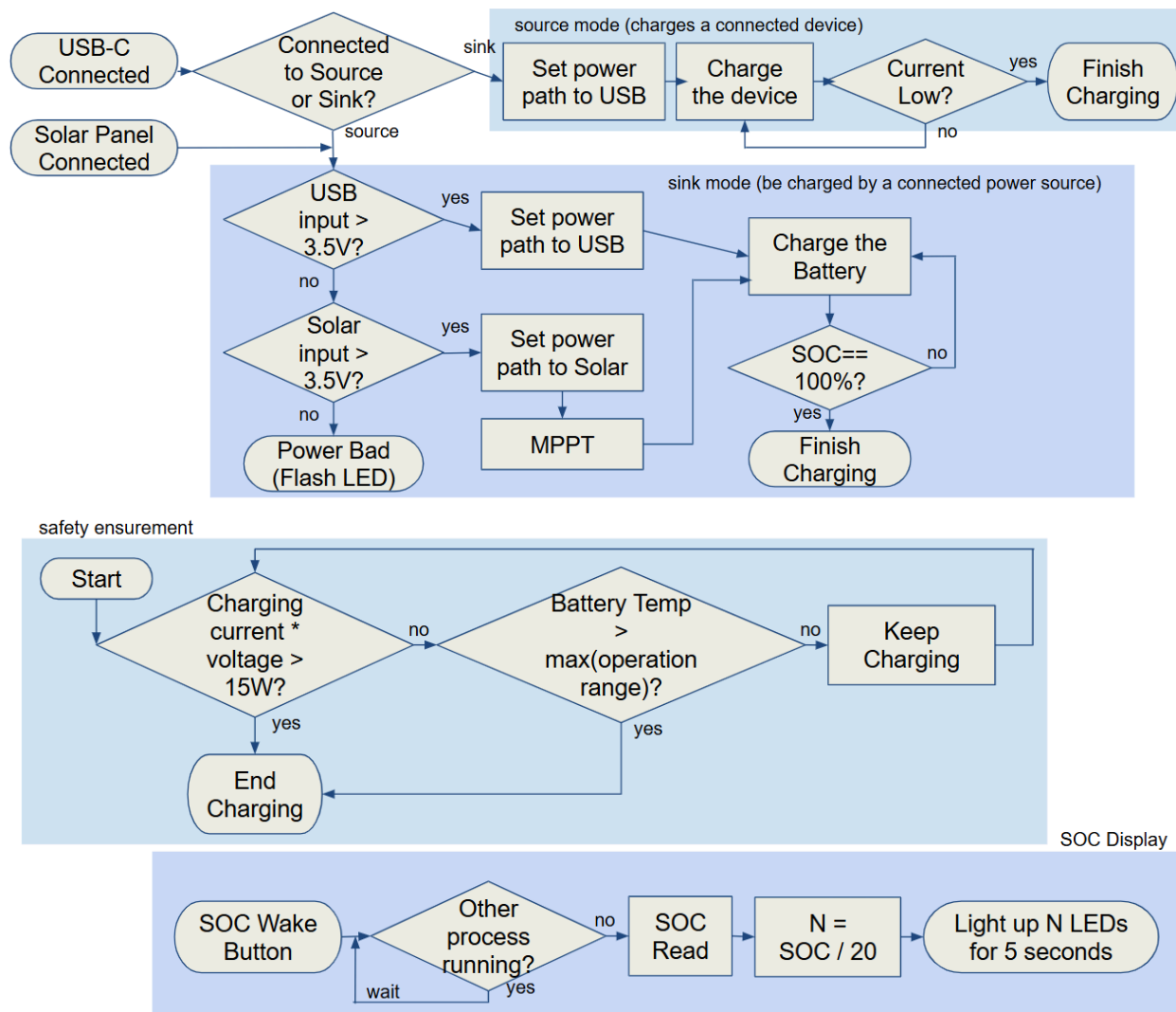


Figure 8: Flowchart of the MCU firmware

2 Integration Design

The design of the system board combines the board designs of the power mux, battery charger, battery gauge, mcu, and USB-C subsystem. The system board is illustrated below in Figure 7, featuring four layers, 50 mil traces between the power paths of the gauge, charger, and mux. The MCU is placed away from the power paths and is able to disconnect from the rest of the system via three shunts to ensure system quality and easier debugging. Additionally, many test points and pin headers were added to the final board to allow for ease of testing and validation. The board is 160.66 by 94.23 mm which is within the allocated size specified by the FSR.

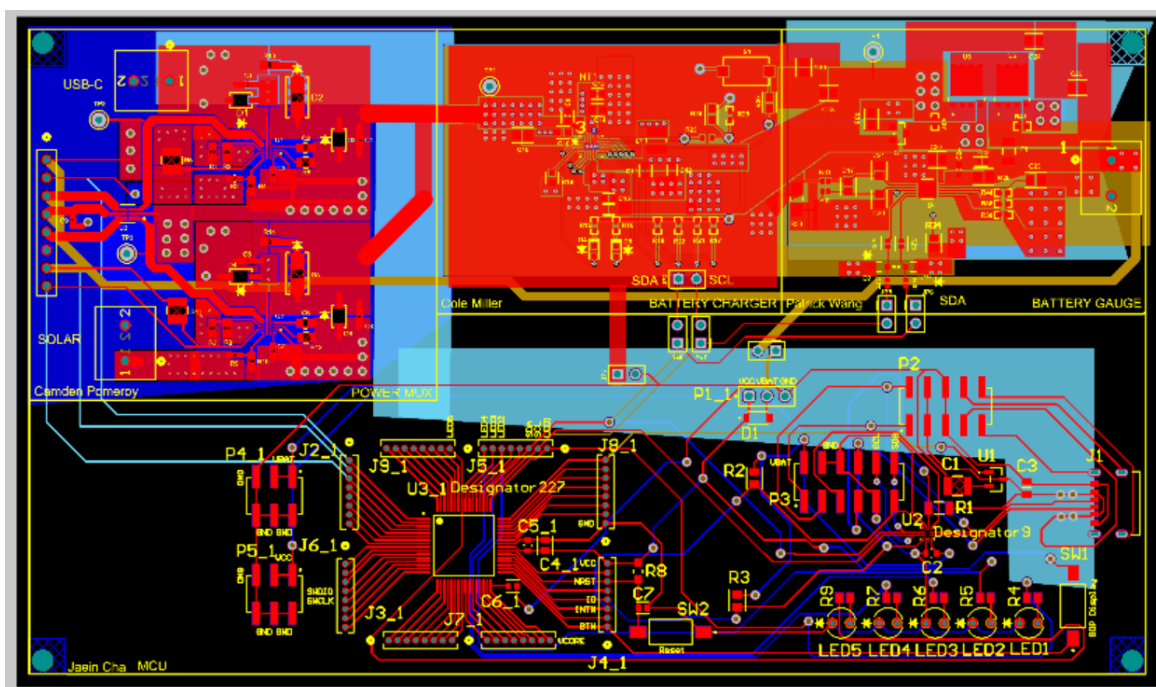


Figure 9: PCB layout of Integrated System

The custom-made 3D-printed box, shown in Figure 8, contains the board measuring $9.5 \times 4 \times 2.5$ inches thus adhering to FSR size requirements. A sturdy mount of the board rests on four standouts inside the enclosure and remains in place during use and transit through the use of four 6-32 screws.

The State of Charge (SOC) information appears through five LEDs in the bottom right section of the board, shown in Figure 9. The battery condition can be checked by observing the five LEDs on the grid which shows full charge when all lights are on and empty condition when no lights show. Users can understand battery health by simply looking at the simple LED indications which avoids requiring external measurement equipment.

Figure 8 shows the entire system fully set up, with all necessary accessories plugged in. Figure 9 illustrates the fully integrated board including all subsystems, and Table 1 labels all of the different subsystems on the integrated board.



Figure 10: Enclosure with Solar Panel and USB-C input

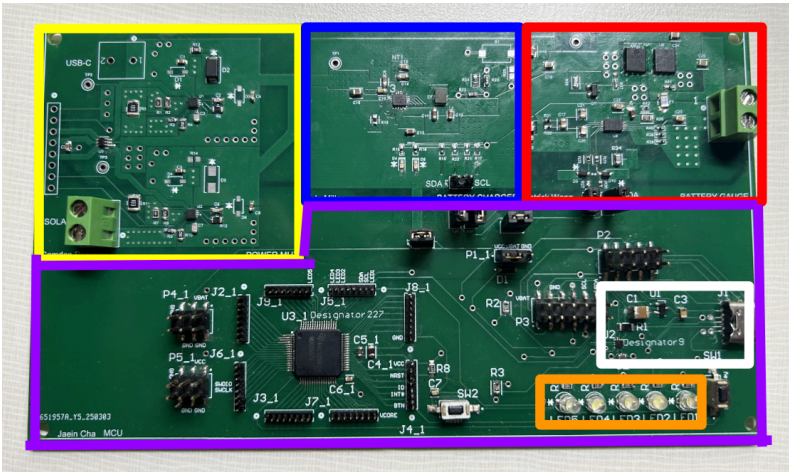


Figure 11: Integrated Board with Individual Subsystems

Color	Subsystem
Yellow	Power Mux
Blue	Battery charger
Red	Battery Gauge
Purple	MCU
Orange	LED SOC Display
White	USB

Table 1: Division of Integrated System

3 System Validation

Following our subsystem validations it was important to test how well things integrated and interacted with each other. We performed multiple system tests and fully validated the Solar Battery pack. We Followed our validation plan below and have included screenshots of data and the results of our product.

Number	Validation Test	Input	Expected Output	Results	Status	Engineer	Reference Number
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	As expected		Camden Pomeroy	3.2.3.2.2
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected		Camden Pomeroy	3.2.3.2.2
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	As expected		Camden Pomeroy	6.1.1
4	PP1 PG (Bad signal)	If Vin < 2V	PG signal is read High from MCU	As expected		Camden Pomeroy	6.1.1
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	As expected		Camden Pomeroy	6.1.1
6	PP2 PG (Bad signal)	If Vin < 2V	PG signal is read High from MCU	As expected		Camden Pomeroy	6.1.1
7	PP1 UVLO Operational	If Vin < UVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
8	PP2 UVLO Operational	If Vin < UVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
9	PP1 OVLO Operational	If Vin > OVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
10	PP2 OVLO Operational	If Vin > OVLO	No output on VBUS	As expected		Camden Pomeroy	3.2.3.1
11	SEL USBC	SEL is sent low from MCU	PP1 is active and PP2 is not	As expected		Camden Pomeroy	6.1.4
12	SEL Solar	SEL is sent high from MCU	PP2 is active and PP1 is not	As expected		Camden Pomeroy	6.1.4
13	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected		Camden Pomeroy	3.2.3.2.2
14	SOC Display	SOC from 0-100%	1,2,3,4, or 5 LEDs on as expected	Not tested		Jaerin Cha	3.2.3.2.1
15	I2C functionality (1:1, EVM)	Run I2C test code	It reads the default register value	As expected		Jaerin Cha	6.1
16	I2C functionality (1:1, on-board)	Run I2C test code	It reads the default register value	Not tested		Jaerin Cha	6.1
17	I2C functionality (1:N, EVM)	Run I2C test code	It reads the default register values	As expected		Jaerin Cha	6.1
18	I2C functionality (1:N, on-board)	Run I2C test code	It reads the default register values	Not tested		Jaerin Cha	6.1
19	I2C - initialization	3.3V Power supply to the pc	Setup the communication correctly	As expected		Jaerin Cha	6.1
20	I2C - DFP configuration	DFP plug is connected to th	Read TUSB I2C address and valu	As expected		Jaerin Cha	6.1
21	I2C - UFP configuration	UFP plug is connected to th	Read TUSB I2C address and valu	As expected		Jaerin Cha	6.1
22	I2C - DRP configuration	DRP plug is connected to th	Read TUSB I2C address and valu	As expected		Jaerin Cha	6.1
23	Reset	Reset button pressed	MCU resets and stop working whi	As expected		Jaerin Cha	3.2.3.2.1
24	MPPT (simulation)	Various voltages	maintain the power as close as 15	As expected		Jaerin Cha	6.2
25	MPPT (on-board)	Various voltages	maintain the power as close as 15	Not tested		Jaerin Cha	6.2
26	SOP Display	SOP button	Lights up all the LEDs when SOP	Not tested		Jaerin Cha	3.2.3.2.1
27	Standard Operation	5V 3A on VBUS	around 5V 3A on VBUS	4.2V 3.2A on VBAT		Cole Miller	2.1
28	USB OTG Function	4.2V 3.2A on VBAT	around 4.2V 3.2A on VBAT	5V 1.5A on VBUS		Cole Miller	2.2
29	ILIM Register Functionality	If input current >1A	output current >1A	Output 1A		Cole Miller	2.3
30	VLM Functionality	If Vin > 4.2V	Vo > 4.2V	Output 4.2V		Cole Miller	2.4
31	Solar Panel VOC	Solar panel Voc < 18V	Solar panel Voc < 18V	As expected		Cole Miller	6.3.1
32	Battery Charger I2C	Write to registers. Write cha	Read from said registers	As expected		Cole Miller	6.1.2
33	Solar Panel Power	regular usage in sunny day	output 15W	As expected		Cole Miller	6.3.12
34	Battery Gauge I2C	Write to registers. Read batt	Read from said registers	As expected		Patrick Wang	6.1.3
35	Battery Lifespan	Battery connected to load	The lifespan is at least 3 hours	As expected		Patrick Wang	3.2.1.2
36	Power Consumption	Normal Operating Condition	Maximum power consumption is t	As expected		Patrick Wang	3.2.1.2
37	Battery SOH	Discharge batteries from 10	Batteries stay in within an appropi	As expected		Patrick Wang	5.1
38	High Power Path Charge	5V 3A from power MUX	Charges Battery Pack	As expected		Patrick, Cole, Camden	3.2.3.2.2
39	High Power Path Discharge	Load Connected to USBC	Battery Pack Discharges 15W	As expected		Patrick, Cole, Camden	3.2.3.2.2
40	USBC Charging	USBC plug from wall outlet	Battery Pack Charges	As expected		Patrick Wang	6.2.1

Table 2: Solar Battery Pack Validation Plan

You can see in Table 2 that the system is fully validated and the main takeaways from each subsystem are as follows. The power mux continues to ensure the charge/discharge at input/output are in a safe operation with OV/UV, and correctly communicate to the MCU to select the solar panel or USB-C as input. The charger, gauge and USB can also all use I2C to send data to the MCU. The charger works as expected in boost and buck mode. The gauge acts as the battery management system shutting down operation when the batteries are fully charged or discharged. This ensures the safety and reliability of the three lithium ion batteries. The MCU can communicate via I2C to control the current draw and through a MPPT algorithm extract the most power possible from our systems solar panels. The MCU can display the SOC through 5 LEDs and control the logic of the TUSB320 to enable charge/discharge through the actual USB-C port. Below are the results and validation tests performed that met the requirements outlined in FSR and system validation plan.



Figure 12: CC1 and CC2 Communication from TUSB320

Depicted on the left is the battery charger looking for a device connected to the port. Whenever a port is found the clock signal stops repeating and the signal forms a high low pair to demonstrate a device is connected and the orientation of the port as shown on the right image. This demonstrates that the device can recognize when devices are connected.



Figure 13: Power MUX En High and Low Graphs

The Power Mux should only allow for bidirectional power flow when the MCU sends a high signal to enable the power path. On the left we have a high signal on the enable pin which produces about a 5 V output onto the output of the power mux subsystem. And on the right we have the MCU sending a low signal to the enable pin, and no voltage on the output of the power mux.

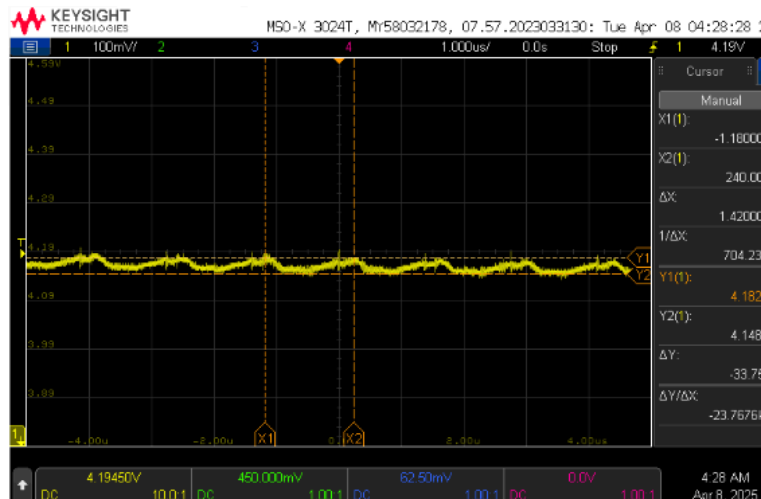
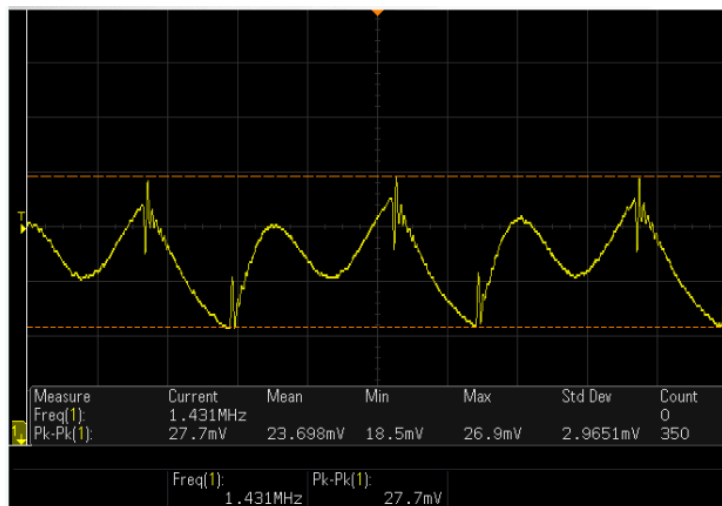


Figure 14: Battery Charger Buck and Boost Waveforms

In order to maximize power efficiency the voltage ripple in both to be as small as possible. Therefore the goal was for there to be a 1% voltage rippler in both modes of operation. Shown on the left is a 30mV ripple while boosting tot 5V DC making a voltage swing of 0.6% voltage swing. On the right is the buck output at 4.2V with a 40mV ripple creating a roughly 1% voltage ripple. Having both of these values below 1% ensured as high efficiency as possible with current components.

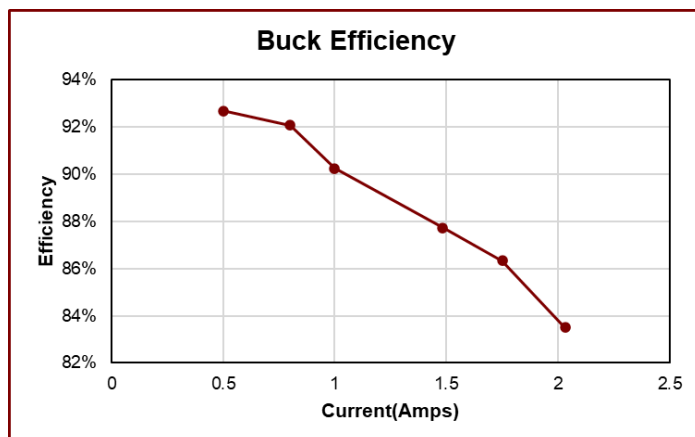
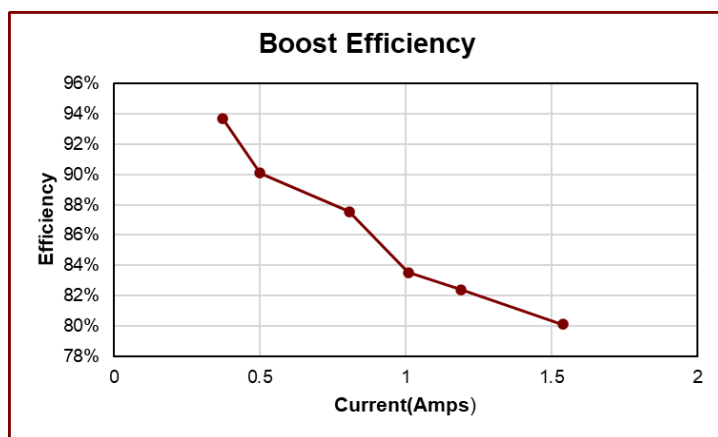


Figure 15: Efficiency of Buck/Boost of Battery Charger

Utilizing the output registers the efficiency was measured as current increased. The test on the left was boosting from 3.7V to 5V and experienced large losses as the output reached close to its maximum of 2A. Then on the right is the device bucking from 5V to 3.7V experiencing similar losses. This is quicker than what the bq25622 is quoted at and while the device uses low ESR components it still experiences large losses. This is due to the heat being generated by the board. Even utilizing large poly pours and parallel paths the board still gets very warm due to the current. The main solution to this would be to use 2 oz pours like TI recommends but that was not in the scope of the project.

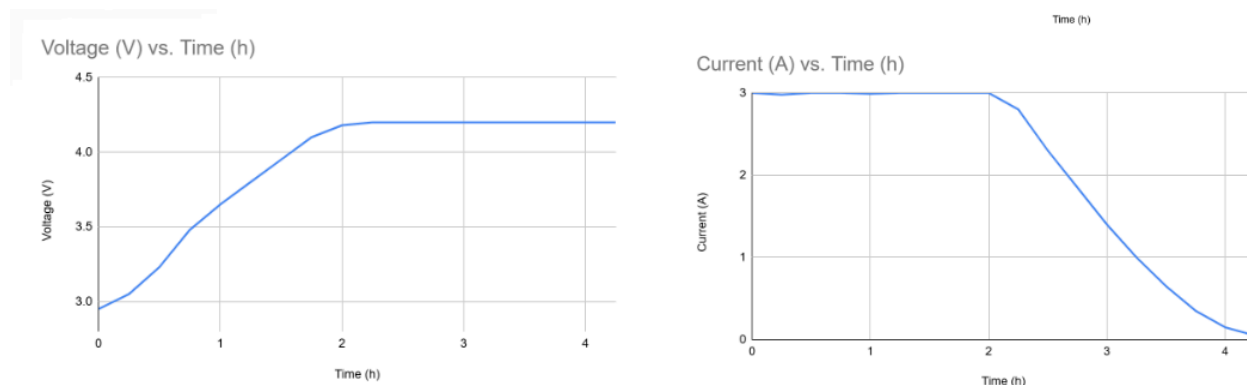


Figure 16: Voltage and Current Vs Time for Charging of Batteries

The figures above illustrate the charging behavior given a 4.2V, 3A input, and follows a standard lithium-ion battery CC-CV behavior within the 1S3P Samsung 18650 30Q battery pack.

During the first phase from 0 until 2 hours the battery operates with a constant 3A current. The voltage increased linearly from 3.0V until it reached 4.2V throughout this charging period. The battery voltage followed a generally linear pattern until it hit a plateau at 4.2V, while the charging current rate stayed at its peak until about 2 hours. The battery reaches its 70–80% full capacity charging point within this rapid charging phase.

The switching of the charging system to the constant voltage phase occurs when the battery reaches its target voltage of 4.2V. The system sustains a 4.2V constant voltage during which the current flow diminishes according to the right-hand graphical display. Between 2 and 4 hours the charging current transitions from its initial value at 3A to close zero Amps. The battery maintains protection through this behavior because it reduces stress during last-stage charging without exposing the battery to excessive current or voltage which extends service life.

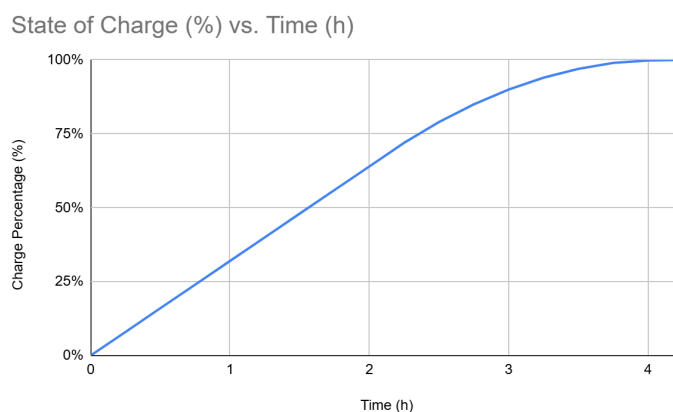


Figure 17: SOC Vs Time for Charging of Batteries

The figure above illustrates the state of charge of the batteries as they charge until 100%. The curve roughly follows that of the Voltage vs. Time graph, and takes just over 4 hours and 20 minutes to fully charge.

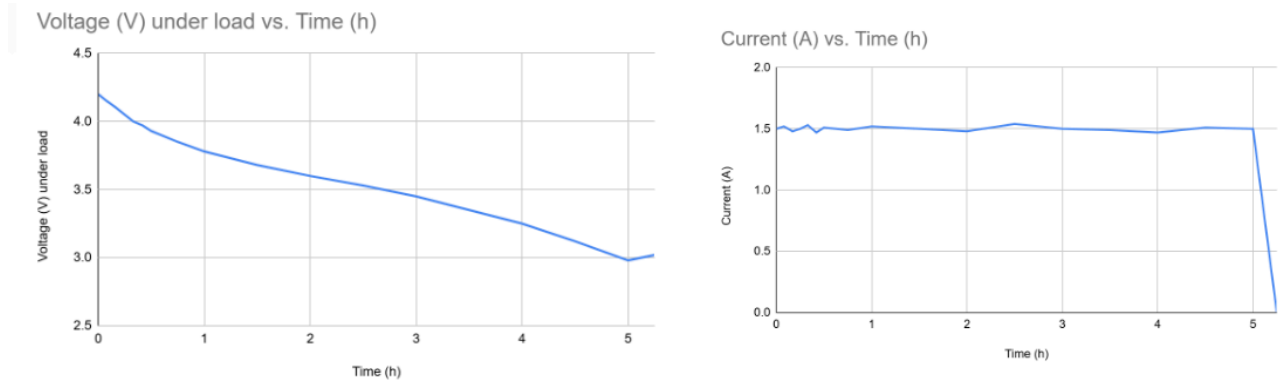


Figure 18: Voltage and Current Vs Time for Discharging of Batteries

The figures above illustrate the discharging behavior of the battery pack given a 1.5A load.

The battery showed a steady voltage drop in the period from 4.2V to 3.0V across 5 hours according to the left graph. The gradual decrease displays the common discharge pattern of lithium-ion cells which produce a straight linear voltage drop in most usage time. When reaching near 3.0V voltage the slope becomes steeper because the battery moves toward complete depletion. Under those operating conditions the battery chemistry faces difficulties in keeping its voltage stable after the state of charge drops to about 20 percent.

The discharge current displays steady operation at 1.5A through the entire discharge time (right graph). Small variable changes exist yet they remain minuscule which suggests stable operation of both load and system components. The current reaches zero immediately following the 5-hour mark which indicates that either the battery reaches complete discharge state or the system stops the load to safeguard the battery from excessive depletion. The provided plots show a healthy battery response with a proper system cutoff when the voltage limit reaches 3.0V.

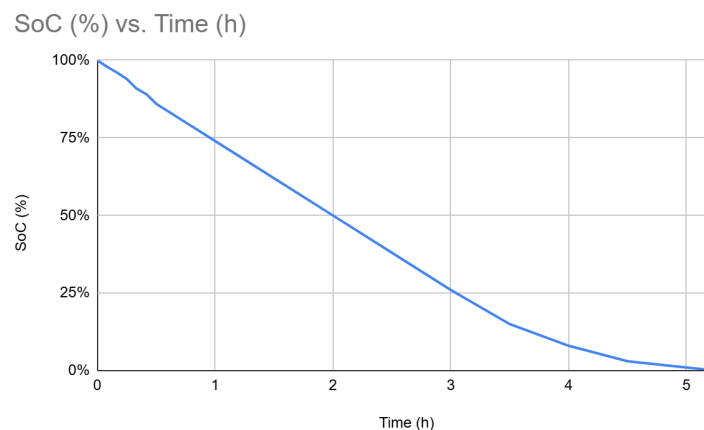


Figure 19: SOC Vs Time for Discharging of Batteries

As seen in the above figure, the state of charge graph mimics that of the voltage vs. time graph quite closely.

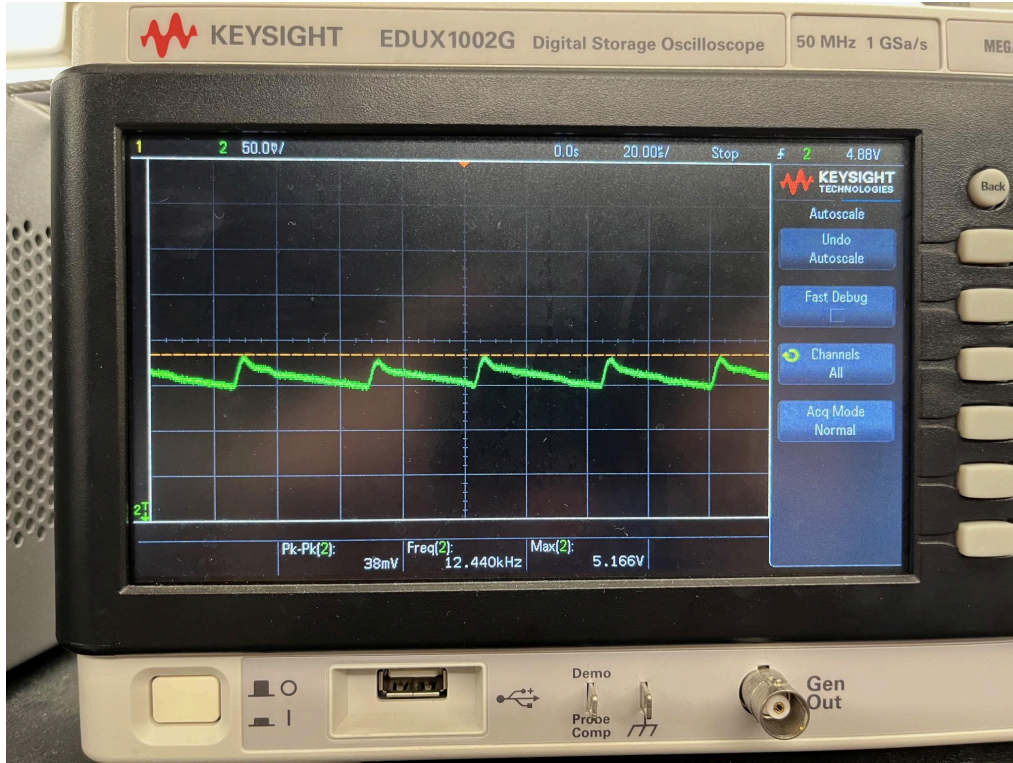


Figure 20: Time Waveform of Charge on USB-C Port

After plugging in a USB-C device such as a phone the waveform above was produced. This is about 5V at 3A which meets the charging standard of USB-C and the specs outlined in the FSR.

Device	Charge Time
USB-C (5V 3A)	2 hrs 45 min
Solar Sunny (7V 2A)	3 hr
Solar Cloudy (7V 0.5A)	10 hrs 30 min

Table 3: Charging Results Given Different Input Conditions

When charging the battery pack it will take about 3 hours if the conditions are a USB-C input or using the solar panels in sunny weather. If cloudy outside the solar panels will take

about ten and a half hours to complete charging. On a normal day there will be some mix of sun and cloud which will result in the batteries charging somewhere in between realistically.

Mode	Current	Discharge Time
USB 2.0	0.5 A	16 hr 40 min
USB 3.0 / 3.1	0.9 A	9 hr
USB BC 1.2	1.5 A	5 hr 15 min
USB-C Default	3 A	2 hr 30 min

Table 4: Discharging Results Given Different Input Conditions

When fast charging a device the battery pack lasts about 2 and a half hours to discharge. This is due to the efficiency in boost mode and the 9000 mAh of the batteries. This is within the design requirements.

System Operating Mode	Current Consumption
Normal	1.2795 mA
Sleep	547.1 μ A
Off	2.1 μ A

Table 5: Current Consumption of System

The current consumption of the system in sleep mode is 547.1 microamps. This concludes that the batteries will last 16450 hours or about 685 days when the device is idle in sleep mode. It is important to the user that the shelf life of the device when not in use remains high.

4 Conclusion

While everything was validated and the project did work, there were some issues with the final devices. Without larger pours the device experiences large heat losses and struggles keeping up 3A. The device also struggles to determine what to do when it sees another dual-role port.

If improvements were to be made to the project some suggestions would be to increase the battery chargers maximum input voltage. This would allow a large amount of solar panels to be used and other USB-C chargers that may be able to get to 20V. Increase the pours to 2 oz to properly dissipate heat. Then lastly change the TUSB320 to the TUSB322 as it has the ability to bias as a sink or a source to interact with other dual-role ports.

However, TI thought the project was acceptable. They are utilizing the design as a reference document for a new IC they're bringing out that is the combination of the PMUX, charger, and TUSB320.

In conclusion, the current design achieves the goal of the project by providing a user an ability to charge their small devices in remote settings and charging the battery pack itself using either USB-C input or solar panels, however, there are various ways to improve this prototype of solar powered battery pack via changing operation range and using different ICs that has improved functionality and works better with the minor problems of the current version of this prototype has.